

The Implementation of the caption package*

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Abstract

The caption package consists of two parts – the kernel (`caption3.sty`) and the main package (`caption.sty`).

The kernel provides all the user commands and internal macros which are necessary for typesetting captions and setting parameters regarding these. While the standard $\text{\LaTeX}$ document classes provide an internal command called `\@makecaption` and no options to control its behavior (except the vertical skips above and below the caption itself), we provide similar commands called `\caption@make` and `\caption@@make`, but with a lot of options which can be selected with `\captionsetup`. Loading the kernel part do not change the output of a $\text{\LaTeX}$ document – it just provides functionality which can be used by $\text{\LaTeX 2}_\epsilon$ packages which typesets captions, for example the caption and subfig packages.

The caption package itself redefines the $\text{\LaTeX}$ commands `\caption`, `\@caption`, and `\@makecaption` and maps the latter one to `\caption@@make`, giving the user the possibility to control the look & feel of the captions from floating environments like `figure` and `table`. Furthermore it does similar to the caption stuff coming from other packages (like the `longtable` or `supertabular` package): Mapping the appropriate internal commands (like `\LT@makecaption` or `\ST@caption`) to the ones offered by the `caption3` kernel. So you can think of the caption package as a layer package, it simply provides adaptation layers between the caption stuff coming from $\text{\LaTeX 2}_\epsilon$ or packages, and the caption stuff offered by the `caption3` kernel.

User manuals

This document is describing the code implementation only. The user documentation can be found in

caption-eng.pdf	The English documentation
caption-rus.pdf	The Russian documentation ¹
caption-deu.pdf	The German documentation

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¹Thanks a lot to Olga Lapko for this translation

Contents

1 Kernel

1.1 Identification

```
1 \NeedsTeXFormat{LaTeX2e}[1994/12/01]
2 \ProvidesPackage{caption3}[2007/11/04 v3.1e caption3 kernel (AR)]
```

1.2 Generic helpers

`\@nameundef` This is the opposite to `\@namedef` which is offered by the $\text{\LaTeX}$ kernel. We use it to remove the definition of some commands and keyval options after `\begin{document}` (to save $\text{\TeX}$ memory) and to remove caption options defined with `\captionsetup[⟨type⟩]`.

```
3 \providecommand*\@nameundef[1]{%
4   \expandafter\let\csname #1\endcsname\@undefined}
```

`\l@addto@macro` The $\text{\LaTeX}$ 2 ϵ kernel offers the internal helper macro `\g@addto@macro` which globally adds tokens to existing macros, like in `\AtBeginDocument`. This is the same but it works local, not global (using `\edef` instead of `\xdef`).

```
5 \providecommand\l@addto@macro[2]{%
6   \begingroup
7     \toks@\expandafter{#1#2}%
8     \edef\@tempa{\endgroup\def\noexpand#1{\the\toks@}}%
9   \@tempa}
```

`\bothIfFirst` `\bothIfFirst` tests if the first argument is not empty, `\bothIfSecond` tests if the second argument is not empty. If yes both arguments get typeset, otherwise none of them.

```
10 \def\bothIfFirst#1#2{%
11   \protected@edef\caption@tempa{#1}%
12   \ifx\caption@tempa\@empty \else
13     #1#2%
14   \fi}
15 \def\bothIfSecond#1#2{%
16   \protected@edef\caption@tempa{#2}%
17   \ifx\caption@tempa\@empty \else
18     #1#2%
19   \fi}
```

`\caption@ifinlist` This helper macro checks if the first argument is in the comma separated list which is offered as second argument. So for example

```
\caption@ifinlist{frank}{axel,frank,olga,steven}{yes}{no}
```

would expand to yes.

```
20 \newcommand*\caption@ifinlist{%
21   \@expandtwoargs\caption@@ifinlist}
22 \newcommand*\caption@@ifinlist[2]{%
23   \begingroup
24   \def\@tempa##1,##2\@nil{%
25     \endgroup
```

```

26     \ifx\relax##2\relax
27         \expandafter\@secondoftwo
28     \else
29         \expandafter\@firstoftwo
30     \fi}%
31     \@tempa,#2,#1,\@nil}%

```

```

\caption@ifin@list \caption@ifin@list{\<cmd>}{\<list entry>}{\<yes>}{\<no>}

```

```

32 \newcommand*\caption@ifin@list[2]{%
33     \caption@ifempty@list#1%
34     {\@secondoftwo}%
35     {\@expandtwoargs\caption@@ifinlist{#2}{#1}}}

```

```

\caption@g@addto@list \caption@g@addto@list{\<cmd>}{\<list entry>}
36 \newcommand*\caption@g@addto@list[2]{%
37     \caption@ifempty@list#1{\gdef#1{#2}}{\g@addto@macro#1{,#2}}}

```

```

\caption@l@addto@list \caption@l@addto@list{\<cmd>}{\<list entry>}
38 \newcommand*\caption@l@addto@list[2]{%
39     \caption@ifempty@list#1{\def#1{#2}}{\l@addto@macro#1{,#2}}}

```

```

\caption@g@removefrom@list \caption@g@removefrom@list{\<cmd>}{\<list entry>}
40 \newcommand*\caption@g@removefrom@list[2]{%
41     \caption@l@removefrom@list#1{#2}%
42     \global\let#1#1}

```

```

\caption@l@removefrom@list \caption@l@removefrom@list{\<cmd>}{\<list entry>}
Caveat: <cmd> will be expanded during this process since \@removeelement is using \edef
to build the new list!
43 \newcommand*\caption@l@removefrom@list[2]{%
44     \caption@ifempty@list#1{\@expandtwoargs\@removeelement{#2}#1#1}}

```

```

\caption@for@list \caption@for@list{\<cmd>}{\<code with #1>}
45 \newcommand*\caption@for@list[2]{%
46     \caption@ifempty@list#1{}{%
47         \def\caption@tempb##1{#2}%
48         \@for\caption@tempa:=#1\do{%
49             \expandafter\caption@tempb\expandafter{\caption@tempa}}}}

```

```

\caption@ifempty@list \caption@ifempty@list{\<cmd>}{\<true>}{\<false>}
50 \newcommand*\caption@ifempty@list[1]{%
51     \ifx#1\@undefined
52         \expandafter\@firstoftwo
53     \else\ifx#1\relax
54         \expandafter\expandafter\expandafter\@firstoftwo
55     \else\ifx#1\@empty
56         \expandafter\expandafter\expandafter\expandafter
57         \expandafter\expandafter\expandafter\@firstoftwo
58     \else

```

```

59 \expandafter\expandafter\expandafter\expandafter
60 \expandafter\expandafter\expandafter\@secondoftwo
61 \fi\fi\fi}

```

For setting and testing boolean options we offer these three helper macros:

```

\caption@setbool
\caption@set@bool
\caption@ifbool
\caption@undefbool

\caption@setbool{<name>}{<value>}
      (with value = false/true/no/yes/off/on/0/1)
\caption@ifbool{<name>}{<if-clause>}{<else-clause>}
\caption@undefbool{<name>}

62 \newcommand*\caption@setbool[1]{%
63 \expandafter\caption@set@bool\csname caption@if#1\endcsname}

64 \newcommand*\caption@set@bool[2]{%
65 \caption@ifinlist{#2}{1,true,yes,on}{%
66 \let#1\@firstoftwo
67 }{\caption@ifinlist{#2}{0,false,no,off}{%
68 \let#1\@secondoftwo
69 }{%
70 \caption@Error{Undefined boolean value `#2'}%
71 }}}

72 \newcommand*\caption@ifbool[1]{\@nameuse{caption@if#1}}
73 \newcommand*\caption@undefbool[1]{\@nameundef{caption@if#1}}

\caption@teststar
\caption@teststar{<cmd>}{<star arg>}{<non-star arg>}
\caption@teststar@{<cmd>}{<star arg/token>}{<non-star arg/token>}
74 \newcommand*\caption@teststar[3]{\@ifstar{#1{#2}}{#1{#3}}}

75 \newcommand*\caption@teststar@[3]{%
76 \@ifstar{#1{#2}}{\caption@ifatletter{#1{#2}}{#1{#3}}}}

77 \newcommand*\caption@ifatletter{%
78 \ifnum\the\catcode`\@=11
79 \expandafter\@firstoftwo
80 \else
81 \expandafter\@secondoftwo
82 \fi}

\caption@withoptargs
\caption@withoptargs{<cmd>}

83 \newcommand*\caption@withoptargs[1]{%
84 \@ifstar
85 {\def\caption@tempa{*}\caption@@withoptargs#1}%
86 {\def\caption@tempa{}\caption@@withoptargs#1}}

87 \def\caption@@withoptargs#1{%
88 \@ifnextchar[%
89 {\caption@@@withoptargs#1}%
90 {\caption@@@@withoptargs#1}}

```

```

91 \def\caption@@@withoptargs#1[#2]{%
92   \l@addto@macro\caption@tempa{[#{#2}]}%
93   \caption@@@withoptargs#1}

94 \def\caption@@@withoptargs#1{%
95   \expandafter#1\expandafter{\caption@tempa}}

\caption@CheckCommand \caption@CheckCommand{<macro>}{<definition of macro>}
\caption@IfCheckCommand checks if a command already exists, with the same definition. It can be used more-than-
once to check if one of multiple definitions will finally match. (It redefines itself later on
to \@gobbletwo if the two commands match fine, making further checks harmless.)
\caption@IfCheckCommand{<true>}{<false>}
will execute the <true> code if one match was finally given, the <false> code otherwise.
(It simply checks if \caption@CheckCommand is \@gobbletwo and restores the
starting definition of \caption@CheckCommand.)

96 \newcommand\caption@DoCheckCommand[2]{%
97   \begingroup
98     \let\@tempa#1%
99     #2%
100    \ifx\@tempa#1%
101      \endgroup
102      \let\caption@CheckCommand\@gobbletwo
103    \else
104      \endgroup
105    \fi}
106 \@onlypreamble\caption@DoCheckCommand

107 \let\caption@CheckCommand\caption@DoCheckCommand
108 \@onlypreamble\caption@CheckCommand

109 \newcommand*\caption@IfCheckCommand{%
110   \ifx\caption@CheckCommand\@gobbletwo
111     \let\next\@firstoftwo
112   \else
113     \let\next\@secondoftwo
114   \fi
115   \let\caption@CheckCommand\caption@DoCheckCommand
116   \next}
117 \@onlypreamble\caption@IfCheckCommand

\caption@AtBeginDocument \caption@AtBeginDocument*{<code>}
Same as \AtBeginDocument but the execution of code will be surrounded by two
\PackageInfos. The starred variant causes the code to be executed after all code
specified using the non-starred variant.

118 \let\caption@begindocumenthook\@empty
119 \let\caption@@begindocumenthook\@empty

120 \def\caption@AtBeginDocument{%
121   \caption@teststar\g@addto@macro
122   \caption@@begindocumenthook\caption@begindocumenthook}
123 \@onlypreamble\caption@AtBeginDocument

```

```

124 \AtBeginDocument{%
125   \PackageInfo{caption}{Begin \noexpand\AtBeginDocument code\@gobble}%
126   \def\caption@AtBeginDocument{%
127     \@ifstar{\g@addto@macro\caption@@begindocumenthook}\@firstofone}%
128   \caption@begindocumenthook
129   \let\caption@begindocumenthook\@undefined
130   \def\caption@AtBeginDocument{%
131     \@ifstar\@firstofone\@firstofone}%
132   \caption@@begindocumenthook
133   \let\caption@@begindocumenthook\@undefined
134   \PackageInfo{caption}{End \noexpand\AtBeginDocument code\@gobble}}

```

1.3 Errors

```

\caption@Warning \caption@Error{<message>}
\caption@WarningNoLine 135 \newcommand*\caption@Warning[1]{%
\caption@Error 136 \caption@WarningNoLine{#1\on@line}}
\caption@KV@err 137 \newcommand*\caption@WarningNoLine[1]{%
138 \PackageWarning{caption}{#1.^J\caption@wh\@gobbletwo}}
139 \newcommand*\caption@Error[1]{%
140 \PackageError{caption}{#1}\caption@eh}
141 \let\caption@KV@err\caption@Error

\caption@wh At the moment we only offer these two simple warning resp. error helpers.
\caption@eh 142 \newcommand*\caption@wh{%
143 See the caption package documentation for explanation.}
144 \newcommand*\caption@eh{%
145 If you do not understand this error, please take a closer look\MessageBreak
146 at the documentation of the 'caption' package, especially the\MessageBreak
147 section about errors.\MessageBreak\@ehc}

```

1.4 Using the keyval package

We need the keyval package for option handling, so we load it here.

```
148 \RequirePackage{keyval}[1997/11/10]
```

```

\undefine@key \undefine@key{<family>}{<key>}
This helper macro is the opposite of \define@key, it removes a keyval definition.

```

```

149 \providecommand*\undefine@key[2]{%
150 \@nameundef{KV@#1@#2}\@nameundef{KV@#1@#2@default}}

```

```

\@onlypreamble@key \onlypreamble@key{<family>}{<key>}
Analogous to \@onlypreamble from LATEX 2ε.
151 \providecommand*\@preamble@keys{}
152 \providecommand*\@onlypreamble@key[2]{\@cons\@preamble@keys{{#1}{#2}}}
153 \@onlypreamble\@onlypreamble@key
154 \@onlypreamble\@preamble@keys

```

```

155 \providecommand*\@notprerr@key[1]{\KV@err{Can be used only in preamble}}
156 \caption@AtBeginDocument*{%
157   \def\@elt#1#2{\expandafter\let\csname KV@#1@#2\endcsname\@notprerr@key}%
158   \@preamble@keys
159   \let\@elt\relax}

\DeclareCaptionOption \DeclareCaptionOption{<option>}[<default value>]{<code>}
\DeclareCaptionOption*{<option>}[<default value>]{<code>}
We declare our options using these commands (instead of using \DeclareOption
offered by LATEX 2ε), so the keyval package is used. The starred form makes the op-
tion available during the lifetime of the current package only, so they can be used with
\usepackage, but not with \captionsetup later on.
160 \newcommand*\DeclareCaptionOption{%
161   \caption@teststar\caption@declareoption\AtEndOfPackage\@gobble}
162 \@onlypreamble\DeclareCaptionOption

163 \newcommand*\caption@declareoption[2]{%
164   #1{\undefine@key{caption}{#2}}\define@key{caption}{#2}}
165 \@onlypreamble\caption@declareoption

DeclareCaptionOptionNoValue \DeclareCaptionOptionNoValue{<option>}{<code>}
\DeclareCaptionOptionNoValue*{<option>}{<code>}
Same as \DeclareCaptionOption but issues an error if a value is given.
166 \newcommand*\DeclareCaptionOptionNoValue{%
167   \caption@teststar\caption@declareoption@novalue\AtEndOfPackage\@gobble}
168 \@onlypreamble\DeclareCaptionOptionNoValue

169 \newcommand\caption@declareoption@novalue[3]{%
170   \caption@declareoption{#1}{#2}[\KV@err]{%
171     \caption@option@novalue{#2}{##1}{#3}}
172 \@onlypreamble\caption@declareoption@novalue

173 \newcommand*\caption@option@novalue[2]{%
174   \ifx\KV@err#2%
175     \expandafter\@firstofone
176   \else
177     \KV@err{No value allowed for #1}%
178     \expandafter\@gobble
179   \fi}

\captionsetup \captionsetup[<type>]{<keyval-list of options>}
\captionsetup*{<type>}{<keyval-list of options>}
If the optional argument ‘type’ is specified, we simply save or append the option list,
otherwise we ‘execute’ it with \setkeys. (The non-starred variant issues a warning if
<keyval-list of options> is not used later on.)
Note: The starred variant will be used inside packages automatically.
180 \newcommand*\captionsetup{%
181   \caption@teststar\@captionsetup\@gobble\@firstofone}

182 \newcommand*\@captionsetup[1]{%
183   \@ifnextchar[\caption@setup@options#1]\caption@setup}

```

```

184 \newcommand*\caption@setup{\caption@setkeys{caption}}
185 \def\caption@setup@options#1[#2]{%
186   % #1 is \@gobble in star form, and \@firstofone otherwise
187   #1{\caption@addtooptlist{#2}}%
188   \expandafter\caption@l@addto@list\csname caption@opt@#2\endcsname}

\clearcaptionsetup \clearcaptionsetup[\langle option \rangle]{\langle type \rangle}
\clearcaptionsetup*[\langle option \rangle]{\langle type \rangle}
This removes the saved option list associated with \langle type \rangle. If \langle option \rangle is given, only this
option will be removed from the list. (The starred variant does not issue warnings.)
Note: The starred variant will be used inside packages automatically.

189 \newcommand*\clearcaptionsetup{%
190   \caption@teststar@\@clearcaptionsetup\@gobble\@firstofone}

191 \newcommand*\@clearcaptionsetup[1]{%
192   \let\caption@tempa#1%
193   \@testopt\@clearcaptionsetup{}}

194 \def\@clearcaptionsetup[#1]#2{%
195   \expandafter\caption@ifempty@list\csname caption@opt@#2\endcsname
196   {\caption@tempa{\caption@Warning{Option list `#2' undefined}}}%
197   {\ifx,#1,%
198     \caption@clearsetup{#2}%
199   \else
200     \caption@@removefromsetup{#1}{#2}%
201   \fi}}

202 \newcommand*\caption@clearsetup[1]{%
203   \caption@removefromoptlist{#1}%
204   \@nameundef{caption@opt@#1}}

205 \newcommand*\caption@removefromsetup{%
206   \let\caption@tempa\@gobble
207   \caption@@removefromsetup}

208 \newcommand*\caption@@removefromsetup[2]{%
209   \expandafter\let\expandafter\@tempa\csname caption@opt@#2\endcsname
210   \expandafter\let\csname caption@opt@#2\endcsname\@undefined
211   \def\@tempb##1=##2\@nil{##1}%
212   \edef\@tempc{#1}%
213   \@for\@tempa:=\@tempa\do{%
214     \edef\@tempd{\expandafter\@tempb\@tempa=\@nil}%
215     \ifx\@tempd\@tempc
216       \let\caption@tempa\@gobble
217     \else
218       \expandafter\expandafter\expandafter\caption@l@addto@list
219       \expandafter\csname caption@opt@#2\expandafter\endcsname
220       \expandafter{\@tempa}%
221     \fi}%
222   \expandafter\caption@ifempty@list\csname caption@opt@#2\endcsname
223   {\caption@removefromoptlist{#2}}}%
224   \caption@tempa{\caption@Warning{%

```

225 Option ‘#1’ was not in list ‘#2’\MessageBreak}}}

\showcaptionsetup \showcaptionsetup[*<package>*]{*<type>*}

This comes for debugging issues: It shows the saved option list which is associated with *<type>*.

```
226 \newcommand*\showcaptionsetup[2][\@firstofone]{%
227   \GenericWarning{ }{%
228     #1 Caption Info: Option list on ‘#2’\MessageBreak
229     #1 Caption Data: \@ifundefined{caption@opt@#2}{%
230       -none-%
231     }{%
232       {\expandafter\expandafter\expandafter\strip@prefix
233        \expandafter\meaning\csname caption@opt@#2\endcsname}%
234     }}}}
```

235 \DeclareCaptionOption{options}{\caption@setoptions{#1}}

\caption@setoptions \caption@setoptions{*<type or environment or...>*}

Caption options which have been saved with \captionsetup[*<type>*] can be executed by using this command. It simply executes the saved option list (and clears it afterwards), if there is any.

```
236 \newcommand*\caption@setoptions[1]{%
237   \caption@Debug{options=#1}%
238   \expandafter\let\expandafter\@tempa\csname caption@opt@#1\endcsname
239   \ifx\@tempa\relax \else
240     \let\caption@iftypewarning\@firstoftwo % = true
241     \expandafter\caption@setup\expandafter{\@tempa}%
242     \caption@clearsetup{#1}%
243   \fi}
```

\caption@addtooptlist \caption@addtooptlist{*<type>*}

\caption@removefromoptlist \caption@removefromoptlist{*<type>*}

Adds or removes an *<type>* to the list of unused caption options. Note that the catcodes of *<type>* are sanitized here so removing *<type>* from the list do not fail when the float package is used (since \float@getstyle gives a result which tokens have catcode 12 = “other”).

```
244 \newcommand*\caption@addtooptlist[1]{%
245   \@ifundefined{caption@opt@#1@lineno}{%
246     \caption@dooptlist\caption@g@addto@list{#1}%
247     \expandafter\xdef\csname caption@opt@#1@lineno\endcsname{\the\inputlineno}%
248   }{}}

249 \newcommand*\caption@removefromoptlist[1]{%
250   \caption@dooptlist\caption@g@removefrom@list{#1}%
251   \global\expandafter\let\csname caption@opt@#1@lineno\endcsname\@undefined}

252 \newcommand*\caption@dooptlist[2]{%
253   \begingroup
254     \edef\@tempa{#2}\@onelevel@sanitize\@tempa
255     \expandafter#1\expandafter\caption@optlist\expandafter{\@tempa}%
256   \endgroup}
```

```

257 \AtEndDocument{%
258   \caption@for@list\caption@optlist{%
259     \PackageWarningNoLine{caption}{%
260       Unused \string\captionsetup[#1]
261       on input line \csname caption@opt@#1@lineno\endcsname}}}%

```

`\caption@setkeys` `\caption@setkeys[<package>]{<family>}{<key-values>}`
 This one simply calls `\setkeys{<family>}{<key-values>}` but lets the error messages not refer to the `keyval` package, but to the *<package>* package instead.

```

262 \newcommand*\caption@setkeys{\@dblarg\caption@@setkeys}
263 \long\def\caption@@setkeys[#1]#2#3{%
264   \expandafter\let\csname ORI@KV@err\caption@keydepth\endcsname\KV@err
265   \expandafter\let\csname ORI@KV@errx\caption@keydepth\endcsname\KV@errx
266   \@ifundefined{#1@KV@err}%
267     {\let\KV@err\ORI@KV@err\let\KV@errx\ORI@KV@errx}%
268     {\expandafter\let\expandafter\KV@err\csname #1@KV@err\endcsname
269      \let\KV@errx\KV@err}%
270   \edef\caption@keydepth{\caption@keydepth i}%
271   \caption@Debug{\protect\setkeys{#2}{#3}}%
272   \setkeys{#2}{#3}%
273   \edef\caption@keydepth{\expandafter\@cdr\caption@keydepth\@nil}%
274   \expandafter\let\expandafter\KV@err\csname ORI@KV@err\caption@keydepth\endcsname
275   \expandafter\let\expandafter\KV@errx\csname ORI@KV@errx\caption@keydepth\endcsname
276 \let\caption@keydepth\@empty

```

`\caption@ExecuteOptions` `\caption@ExecuteOptions{<family>}{<key-values>}`
 We execute our options using the `keyval` interface, so we use this one instead of `\ExecuteOptions` offered by `LATEX 2ε`.

```

277 \newcommand*\caption@ExecuteOptions[2]{%
278   \@expandtwoargs\caption@setkeys{#1}{#2}}%
279 \@onlypreamble\caption@ExecuteOptions

```

`\caption@ProcessOptions` `\caption@ProcessOptions*{<family>}`
 We process our options using the `keyval` package, so we use this one instead of `\ProcessOptions` offered by `LATEX 2ε`. The starred variant do not process the global options. (This code was taken from the `hyperref` package[?] v6.74 and improved.)

```

280 \newcommand*\caption@ProcessOptions{%
281   \caption@teststar\caption@@ProcessOptions\@gobble\@firstofone}
282 \@onlypreamble\caption@ProcessOptions
283 \newcommand*\caption@@ProcessOptions[2]{%
284   \let\@tempc\relax
285   \let\caption@tempa\@empty
286   #1{% \@firstofone -or- \@gobble
287     \@for\CurrentOption:=\@classoptionslist\do{%
288       \@ifundefined{KV@#2@\CurrentOption}{}{%
289         \@ifundefined{KV@#2@\CurrentOption @default}{%
290           \PackageInfo{#2}{Global option '\CurrentOption' ignored}%

```

```

291      }{%
292      \PackageInfo{#2}{Global option '\CurrentOption' processed}%
293      \edef\caption@tempa{\caption@tempa,\CurrentOption,}%
294      \@expandtwoargs\@removeelement\CurrentOption
295      \@unusedoptionlist\@unusedoptionlist
296      }%
297  }%
298  }%
299  \let\CurrentOption\@empty
300  }%
301  \caption@ExecuteOptions{#2}{\caption@tempa\@optionlist{\@currname.\@current}}%
302  \AtEndOfPackage{\let\@unprocessedoptions\relax}%
303  \@onlypreamble\caption@@ProcessOptions

```

1.5 Margin resp. width

`\captionmargin` and `\captionwidth` contain the extra margin resp. the total width used for captions. Please never set these values in a direct way, they are just accessible in user documents to provide compatibility to `caption.sty v1.x`.

Note that we can only set one value at a time, ‘margin’ or ‘width’. If `\captionwidth` is not zero we will take this value afterwards, otherwise `\captionmargin` and `\captionmargin@`.

```

304 \newdimen\captionmargin
305 \newdimen\captionmargin@
306 \newdimen\captionwidth

307 \DeclareCaptionOption{margin}{\setcaptionmargin{#1}}
308 \DeclareCaptionOption{margin*}{\setcaptionmargin*{#1}}
309 \DeclareCaptionOption{width}{\setcaptionwidth{#1}}
310 \DeclareCaptionOption{twoside}[1]{\caption@set@bool\caption@iftwoside{#1}}

311 \DeclareCaptionOption{minmargin}{\caption@setoptcmd\caption@minmargin{#1}}
312 \DeclareCaptionOption{maxmargin}{\caption@setoptcmd\caption@maxmargin{#1}}

```

`\setcaptionmargin` `\setcaptionmargin{<amount>}`
 `\setcaptionmargin*{<amount>}`

Please never use this in user documents, it’s just there to provide compatibility to the `caption2` package.

```

313 \newcommand*\setcaptionmargin{%
314   \caption@teststar\caption@setmargin\@gobble\@firstofone}

315 \newcommand*\caption@setmargin[2]{%
316   #1{\captionwidth\z@}%
317   \caption@@setmargin#2,#2,\@nil}

318 \def\caption@@setmargin#1,#2,#3\@nil{%
319   \setlength\captionmargin@{#2}%
320   \setlength\captionmargin{#1}%
321   \addtolength\captionmargin@{-\captionmargin}}

```

`\setcaptionwidth` `\setcaptionwidth{<amount>}`
Please never use this in user documents, it's just there to provide compatibility to the `caption2` package.

```

322 \newcommand*\setcaptionwidth{%
323   \captionmargin\z@
324   \captionmargin@\z@
325   \setlength\captionwidth{

```

`\caption@counter` This counter numbers the captions. At the moment it will be used inside `\caption@ifoddpage` only.

```

326 \newcommand*\caption@thecounter{0}
327 \newcommand*\caption@stepcounter{%
328   \@tempcnta\caption@thecounter
329   \advance\@tempcnta\@ne
330   \xdef\caption@thecounter{\the\@tempcnta}}

```

`\caption@newlabel` This command is a modified version of `\newlabel` from L^AT_EX2_ε. It will be written to the `.aux` file to pass label information from one run to another. (We use it inside `\caption@ifoddpage` and `\caption@ragged`.)

```

331 \newcommand*\caption@newlabel{\@newl@bel{caption@r}}

```

`\caption@thepage` This command is a modified version of `\thepage` from L^AT_EX2_ε. It will be used inside `\caption@ifoddpage` only.

```

332 \newcommand*\caption@thepage{\the\c@page}

```

`\caption@label` This command is a modified version of `\label` from L^AT_EX2_ε. It will be used inside `\caption@ifoddpage` and `\FP@helpNote`.

```

333 \newcommand*\caption@label[1]{%
334   \protected@write\@auxout{\let\caption@thepage\relax}%
335     {\string\caption@newlabel{#1}{\caption@thepage}}}

```

`\caption@pageref` This command is a modified version of `\pageref` from L^AT_EX2_ε. It will be used inside `\caption@ifoddpage` and `\FP@helpNote`.

```

336 \newcommand*\caption@pageref[1]{%
337   \expandafter\ifx\csname caption@r@#1\endcsname\relax
338     \G@refundefinedtrue % => 'There are undefined references.'
339     \caption@Warning{Reference on page \thepage \space undefined}%
340     \else
341       \expandafter\let\expandafter\caption@thepage\csname caption@r@#1\endcsname
342     \fi}

```

`\caption@ifoddpage` At the moment this macro uses an own label...ref mechanism, but an alternative implementation method would be using the `refcount` package[?] and `\ifodd\getpagerefnumber{...}`.
Note: This macro re-defines itself so the `.aux` file will only be used once per group.

```

343 \newcommand*\caption@ifoddpage{%
344   \caption@iftwoside{%
345     \caption@label\caption@thecounter

```

```

346 \caption@pageref\caption@thecounter
347 \ifodd\caption@thepage
348 \let\caption@ifoddpage\@firstoftwo
349 \else
350 \let\caption@ifoddpage\@secondoftwo
351 \fi
352 \let\caption@ifoddpage\@firstoftwo}%
353 \caption@ifoddpage}

\caption@setoptcmd \caption@setoptcmd{<cmd>}{<off-or-value>}
354 \newcommand*\caption@setoptcmd[2]{%
355 \caption@ifinlist{#2}{0,false,no,off}{\let#1\undefined}{\def#1{#2}}}%

```

1.6 Indentions

These are the indentions we support.

```

\caption@indent
\caption@parindent
\caption@hangindent
356 \newdimen\caption@indent
357 \newdimen\caption@parindent
358 \newdimen\caption@hangindent

359 \DeclareCaptionOption{indent}[\leftmargini]{% obsolete!
360 \setlength\caption@indent{#1}}
361 \DeclareCaptionOption{indentation}[\leftmargini]{%
362 \setlength\caption@indent{#1}}
363 \DeclareCaptionOption{parindent}{%
364 \setlength\caption@parindent{#1}}
365 \DeclareCaptionOption{hangindent}{%
366 \setlength\caption@hangindent{#1}}
367 \DeclareCaptionOption{parskip}{%
368 \l@addto@macro\caption@par{\setlength\parskip{#1}}}%

There is an option clash between the KOMA-Script document classes and the caption
kernel, both define the options parindent and parskip but with different meaning.
Furthermore the ones defined by the caption kernel take a value as parameter but the
KOMA-Script ones do not. So we need special versions of the options parindent and
parskip here which determine if a value is given (and therefore should be treated as
our option) or not (and therefore should be ignored by us).2

369 \@ifundefined{scr@caption}{}{%
370 \let\caption@KV@parindent\KV@caption@parindent
371 \DeclareCaptionOption{parindent}[]{%
372 \ifx,#1,%
373 \caption@Debug{Option 'parindent' ignored}%
374 \else
375 \caption@KV@parindent{#1}%
376 \fi}%

```

²This problem was completely solved due a change of `\caption@ProcessOptions` in the `caption` package v3.0j, but we still need this workaround since these options otherwise would still collide with the current version 1.3 of the `subfig` package (Sigh!)

```

377 \let\caption@KV@parskip\KV@caption@parskip
378 \DeclareCaptionOption{parskip}[]{%
379     \ifx,#1,%
380         \caption@Debug{Option 'parskip' ignored}%
381     \else
382         \caption@KV@parskip{#1}%
383     \fi}%
384 }

```

1.7 Styles

```

\DeclareCaptionStyle \DeclareCaptionStyle{<name>}[<single-line-list-of-KV>]{<list-of-KV>}
385 \newcommand*\DeclareCaptionStyle[1]{%
386     \@testopt{\caption@declarestyle{#1}}{}}
387 \@onlypreamble\DeclareCaptionStyle
388 \def\caption@declarestyle#1[#2]#3{%
389     \global\@namedef{caption@sls@#1}{#2}%
390     \global\@namedef{caption@sty@#1}{#3}}
391 \@onlypreamble\caption@declarestyle
392 \DeclareCaptionOption{style}{\caption@setstyle{#1}}
393 \DeclareCaptionOption{singlelinecheck}{\caption@set@bool\caption@ifslc{#1}}

```

```

\caption@setstyle \caption@setstyle{<name>}
\caption@setstyle*{<name>}

```

Selecting a caption style means saving the additional *<single-line-list-of-KV>* (this will be done by `\caption@sls`), resetting the caption options to the base ones (this will be done using `\caption@resetstyle`) and executing the *<list-of-KV>* options (this will be done using `\caption@setup`).

The starred version will give no error message if the given style is not defined.

```

394 \newcommand*\caption@setstyle{%
395     \caption@teststar\caption@@setstyle\@gobble\@firstofone}
396 \newcommand*\caption@@setstyle[2]{%
397     \@ifundefined{caption@sty@#2}%
398     {#1{\caption@Error{Undefined style '#2'}}}%
399     {\expandafter\let\expandafter\caption@sty\csname caption@sty@#2\endcsname
400     \ifx\caption@setstyle@flag\@undefined
401         \let\caption@setstyle@flag\caption@setstyle
402         \let\caption@iftypewarning\@firstoftwo % = true
403         \caption@resetstyle
404         \expandafter\caption@setup\expandafter{\caption@sty}%
405         \let\caption@setstyle@flag\@undefined
406     \else
407         \expandafter\caption@setup\expandafter{\caption@sty}%
408     \fi
409     \expandafter\let\expandafter\caption@sls\csname caption@sls@#2\endcsname
410     \expandafter\caption@l@addto@list\expandafter\caption@opt@singleline
411     \expandafter{\caption@sls}}

```

`\caption@resetstyle` This resets (nearly) all caption options to the base ones. *Note that this does not touch the skips and the positioning!*

```

412 \newcommand*\caption@resetstyle{%
413   \caption@setup{%
414     format=plain,labelformat=default,labelsep=colon,textformat=simple,%
415     justification=justified,font=,labelfont=,textfont=,%
416     margin=0pt,minmargin=0,maxmargin=0,%
417     indent=0pt,parindent=0pt,hangindent=0pt,%
418     singlelinecheck=1,strut=1}%
419   \caption@clearsetup{singleline}}

```

Currently there are two pre-defined styles, called ‘base’ & ‘default’. The first one is a perfect match to the behavior of `\@makecaption` offered by the standard L^AT_EX document classes (and was called ‘default’ in the caption package v3.0), the second one matches the document class actually used.

```

420 \DeclareCaptionStyle{base}[indent=0pt,justification=centering]{}
421 \DeclareCaptionStyle{default}[indent=0pt,justification=centering]{%
422   format=default,labelsep=default,textformat=default,%
423   justification=default,font=default,labelfont=default,textfont=default}

```

1.8 Formats

`\DeclareCaptionFormat` `\DeclareCaptionFormat{<name>}{<code with #1, #2, and #3>}`
`\DeclareCaptionFormat*{<name>}{<code with #1, #2, and #3>}`

The starred form causes the code being typeset in vertical (instead of horizontal) mode, but does not support the `indentation=` option.

```

424 \newcommand*\DeclareCaptionFormat{%
425   \caption@teststar\caption@declareformat\@gobble\@firstofone}
426 \@onlypreamble\DeclareCaptionFormat

427 \newcommand*\caption@declareformat[2]{%
428   \@dblarg{\caption@@declareformat#1{#2}}}%
429 \@onlypreamble\caption@declareformat

430 \long\def\caption@@declareformat#1#2[#3]#4{%
431   \global\expandafter\let\csname caption@ifh@#2\endcsname#1%
432   \global\long\@namedef{caption@slfmt@#2}##1##2##3{#3}%
433   \global\long\@namedef{caption@fmt@#2}##1##2##3{#4}}%
434 \@onlypreamble\caption@@declareformat

435 \DeclareCaptionOption{format}{\caption@setformat{#1}}

```

`\caption@setformat` `\caption@setformat{<name>}`

Selecting a caption format simply means saving the code (in `\caption@fmt`) and if the code should be used in horizontal or vertical mode (`\caption@ifh`).

```

436 \newcommand*\caption@setformat[1]{%
437   \@ifundefined{caption@fmt@#1}%
438   {\caption@Error{Undefined format `#1'}}%
439   {\expandafter\let\expandafter\caption@ifh\csname caption@ifh@#1\endcsname

```

```

440 \expandafter\let\expandafter\caption@slfmt\csname caption@slfmt@#1\endcsname
441 \expandafter\let\expandafter\caption@fmt\csname caption@fmt@#1\endcsname}}

```

There are two pre-defined formats, called ‘plain’ and ‘hang’.

```

442 \DeclareCaptionFormat{plain}{#1#2#3\par}
443 \DeclareCaptionFormat{hang}[#1#2#3\par]{%
444 \caption@ifin@list\caption@lsep@crlist\caption@lsepname
445 {\caption@Error{%
446 The option ‘labelsep=\caption@lsepname’ does not work\MessageBreak
447 with ‘format=hang’}}}%
448 {\@hangfrom{#1#2}%
449 \advance\caption@parindent\hangindent
450 \advance\caption@hangindent\hangindent
451 \caption@@par#3\par}}

```

‘default’ usually maps to ‘plain’.

```

452 \def\caption@fmt@default{\caption@fmt@plain}
453 \def\caption@ifh@default{\caption@ifh@plain}

```

1.9 Label formats

```

\DeclareCaptionLabelFormat \DeclareCaptionLabelFormat{<name>}{<code with #1, #2, and #3>}
454 \newcommand*\DeclareCaptionLabelFormat[2]{%
455 \global\@namedef{caption@lfmt@#1}##1#2##3{#2}}
456 \@onlypreamble\DeclareCaptionLabelFormat

457 \DeclareCaptionOption{labelformat}{\caption@setlabelformat{#1}}

```

```

\caption@setlabelformat \caption@setlabelformat{<name>}

```

Selecting a caption label format simply means saving the code (in \caption@lfmt).

```

458 \newcommand*\caption@setlabelformat[1]{%
459 \@ifundefined{caption@lfmt@#1}%
460 {\caption@Error{Undefined label format ‘#1’}}%
461 {\expandafter\let\expandafter\caption@lfmt\csname caption@lfmt@#1\endcsname}}

```

There are three pre-defined label formats, called ‘empty’, ‘simple’, and ‘parens’.

```

462 \DeclareCaptionLabelFormat{empty}{}
463 \DeclareCaptionLabelFormat{simple}{\bothIfFirst{#1}{\nobreakspace}#2}
464 \DeclareCaptionLabelFormat{parens}{\bothIfFirst{#1}{\nobreakspace}{#2}}

```

‘default’ usually maps to ‘simple’.

```

465 \def\caption@lfmt@default{\caption@lfmt@simple}

```

1.10 Label separators

```

\DeclareCaptionLabelSeparator \DeclareCaptionLabelSeparator{<name>}{<code>}
\DeclareCaptionLabelSeparator*{<name>}{<code>}

```

The starred form causes the label separator to be typeset *without* using \captionlabelfont.

```

466 \newcommand\DeclareCaptionLabelSeparator{%
467   \caption@teststar\caption@declarelabelseparator\@gobble\@firstofone}
468 \@onlypreamble\DeclareCaptionLabelSeparator

469 \newcommand\caption@declarelabelseparator[3]{%
470   \global\expandafter\let\csname caption@iflfl@#2\endcsname#1%
471   \global\long\@namedef{caption@lsep@#2}{#3}%
472   \caption@@declarelabelseparator{#2}#3\\@nil}
473 \@onlypreamble\caption@declarelabelseparator

474 \long\def\caption@@declarelabelseparator#1#2\\#3\@nil{%
475   \def\@tempa{#3}\ifx\@tempa\@empty \else
476     \caption@g@addto@list\caption@lsep@crlist{#1}%
477   \fi}
478 \@onlypreamble\caption@@declarelabelseparator

479 \DeclareCaptionOption{labelsep}{\caption@setlabelseparator{#1}}
480 \DeclareCaptionOption{labelseparator}{\caption@setlabelseparator{#1}}

```

`\caption@setlabelseparator` `\caption@setlabelseparator{<name>}`

Selecting a caption label separator simply means saving the code (in `\caption@lsep`).

```

481 \newcommand*\caption@setlabelseparator[1]{%
482   \@ifundefined{caption@lsep@#1}%
483     {\caption@Error{Undefined label separator `#1'}}%
484     {\edef\caption@lsepname{#1}%
485       \expandafter\let\expandafter\caption@iflfl\csname caption@iflfl@#1\endcsname
486       \expandafter\let\expandafter\caption@lsep\csname caption@lsep@#1\endcsname}}

```

There are seven pre-defined label separators, called ‘none’, ‘colon’, ‘period’, ‘space’, ‘quad’, ‘newline’, and ‘endash’.

```

487 \DeclareCaptionLabelSeparator{none}{}
488 \DeclareCaptionLabelSeparator{colon}{: }
489 \DeclareCaptionLabelSeparator{period}{. }
490 \DeclareCaptionLabelSeparator{space}{ }
491 \DeclareCaptionLabelSeparator*{quad}{\quad}
492 \DeclareCaptionLabelSeparator*{newline}{\\}
493 \DeclareCaptionLabelSeparator*{endash}{\space\textendash\space}

```

‘default’ usually maps to ‘colon’.

```

494 \def\caption@lsep@default{\caption@lsep@colon}
495 \def\caption@iflfl@default{\caption@iflfl@colon}

```

1.11 Text formats

`\DeclareCaptionTextFormat` `\DeclareCaptionTextFormat{<name>}{<code with #1>}`

```

496 \newcommand*\DeclareCaptionTextFormat[2]{%
497   \global\long\@namedef{caption@tfmt@#1}##1{#2}}
498 \@onlypreamble\DeclareCaptionTextFormat

```

```

499 \DeclareCaptionOption{textformat}{\caption@settextformat{#1}}
500 \DeclareCaptionOption{strut}{\caption@set@bool\caption@ifstrut{#1}}

```

`\caption@settextformat` `\caption@settextformat{<name>}`

Selecting a caption text format simply means saving the code (in `\caption@tfmt`).

```
501 \newcommand*\caption@settextformat[1]{%
502   \@ifundefined{caption@tfmt@#1}%
503     {\caption@Error{Undefined text format '#1'}}%
504     {\expandafter\let\expandafter\caption@tfmt\csname caption@tfmt@#1\endcsname}}
```

There are two pre-defined text formats, called ‘simple’ and ‘period’.

```
505 \DeclareCaptionTextFormat{simple}{#1}
506 \DeclareCaptionTextFormat{period}{#1.}
```

‘default’ usually maps to ‘simple’.

```
507 \def\caption@tfmt@default{\caption@tfmt@simple}
```

1.12 Fonts

`\DeclareCaptionFont` `\DeclareCaptionFont{<name>}{<code>}`

```
508 \newcommand*\DeclareCaptionFont[2]{%
509   \define@key{caption@fnt}{#1}[]{\l@addto@macro\caption@fnt{#2}}
510 \@onlypreamble\DeclareCaptionFont
```

`\DeclareDefaultCaptionFont` `\DeclareDefaultCaptionFont{<name>}{<code>}`

```
511 \newcommand*\DeclareDefaultCaptionFont[2]{%
512   \global\@namedef{caption#1@default}{#2}}
513 \@onlypreamble\DeclareDefaultCaptionFont

514 \DeclareCaptionOption{font}{\caption@setfont{font}{#1}}
515 \DeclareDefaultCaptionFont{font}{}

516 \DeclareCaptionOption{labelfont}{\caption@setfont{labelfont}{#1}}
517 \DeclareDefaultCaptionFont{labelfont}{}

518 \DeclareCaptionOption{textfont}{\caption@setfont{textfont}{#1}}
519 \DeclareDefaultCaptionFont{textfont}{}

\caption@setfont    \caption@setfont{<name>}{<keyval-list of names>}
```

Selecting a caption font means saving all the code snippets (in `\caption#1`). Because we use `\setkeys` recursive here we need to do this inside an extra group and collect all the code snippets in `\caption@tempa` first.

```
520 \newcommand*\caption@setfont[2]{%
521   \begingroup
522     \let\caption@fnt\@empty
523     \define@key{caption@fnt}{default}[]{%
524       \l@addto@macro\caption@fnt{\csname caption#1@default\endcsname}}%
525     \caption@setkeys[caption]{caption@fnt}{#2}%
526     \global\let\caption@tempa\caption@fnt
527   \endgroup
528   \expandafter\let\csname caption#1\endcsname\caption@tempa}
```

```

\caption@font \caption@font{<keyval-list of names>}
\caption@font*{<keyval-code>}
Sets the given font, e.g. \caption@font{small, it} is equivalent to \small\itshape.
529 \newcommand*\caption@font{%
530   \caption@teststar\caption@@font\@firstofone
531     {\caption@setkeys[caption]{caption@fnt}}}%
532 \newcommand*\caption@@font[2]{%
533   \begingroup
534   \def\caption@fnt{\endgroup}%
535   #1{#2}%
536   \caption@fnt}

```

These are the pre-defined font code snippets.

```

537 \DeclareCaptionFont{normalcolor}{\normalcolor}
538 \DeclareCaptionFont{color}{\color{#1}}

539 \DeclareCaptionFont{normalfont}{\normalfont}
540 \DeclareCaptionFont{up}{\upshape}
541 \DeclareCaptionFont{it}{\itshape}
542 \DeclareCaptionFont{sl}{\slshape}
543 \DeclareCaptionFont{sc}{\scshape}
544 \DeclareCaptionFont{md}{\mdseries}
545 \DeclareCaptionFont{bf}{\bfseries}
546 \DeclareCaptionFont{rm}{\rmfamily}
547 \DeclareCaptionFont{sf}{\sffamily}
548 \DeclareCaptionFont{tt}{\ttfamily}

549 \DeclareCaptionFont{scriptsize}{\scriptsize}
550 \DeclareCaptionFont{footnotesize}{\footnotesize}
551 \DeclareCaptionFont{small}{\small}
552 \DeclareCaptionFont{normalsize}{\normalsize}
553 \DeclareCaptionFont{large}{\large}
554 \DeclareCaptionFont{Large}{\Large}

555 \DeclareCaptionFont{singlespacing}{\setstretch\setspace@singlespace}% normally 1
556 \DeclareCaptionFont{onehalfspacing}{\onehalfspacing}
557 \DeclareCaptionFont{doublespacing}{\doublespacing}
558 \DeclareCaptionFont{stretch}{\setstretch{#1}}
559 \caption@AtBeginDocument{\providecommand*\setstretch[1]{} }

560 %\DeclareCaptionFont{normal}{%
561 %  \caption@font{normalcolor,normalfont,normalsize,singlespacing}
562 %\DeclareCaptionFont{normal}{%
563   \caption@font*{%
564     \KV@caption@fnt@normalcolor\@unused
565     \KV@caption@fnt@normalfont\@unused
566     \KV@caption@fnt@normalsize\@unused
567     \KV@caption@fnt@singlespacing\@unused}}

```

`\captionsize` The old versions *v1.x* of the caption package offered this command to setup the font size used for captions. We still do so old documents will work fine.

```

568 \providecommand\captionsize{}

```

```

569 \DeclareCaptionOption{size}{\caption@setfont{size}{#1}}
570 \DeclareDefaultCaptionFont{size}{}

```

1.13 Justifications

```

DeclareCaptionJustification \DeclareCaptionJustification{<name>}{<code>}
571 \newcommand*\DeclareCaptionJustification[2]{%
572   \global\@namedef{caption@hj@#1}{#2}% for compatibility to v3.0
573   \DeclareCaptionFont{#1}{#2}}
574 \@onlypreamble\DeclareCaptionJustification

DefaultCaptionJustification \DeclareDefaultCaptionJustification{<code>}
575 \newcommand*\DeclareDefaultCaptionJustification[1]{%
576   \global\@namedef{caption@hj@default}{#1}% for compatibility to v3.0
577   \DeclareDefaultCaptionFont{@hj}{#1}}
578 \@onlypreamble\DeclareDefaultCaptionJustification

579 \DeclareCaptionOption{justification}{\caption@setjustification{#1}}
580 \DeclareDefaultCaptionJustification{}

\caption@setjustification \caption@setjustification{<name>}
Selecting a caption justification simply means saving the code (in \caption@hj).
581 \newcommand*\caption@setjustification{\caption@setfont{@hj}}

```

These are the pre-defined justification code snippets.

```

582 \DeclareCaptionJustification{justified}{}
583 \DeclareCaptionJustification{centering}{\centering}
584 \DeclareCaptionJustification{centerfirst}{\centerfirst}
585 \DeclareCaptionJustification{centerlast}{\centerlast}
586 \DeclareCaptionJustification{raggedleft}{\raggedleft}
587 \DeclareCaptionJustification{raggedright}{\raggedright}

\centerfirst Please blame Frank Mittelbach for the code of \centerfirst :-)
588 \providecommand\centerfirst{%
589   \let\\@centercr
590   \edef\caption@normaladjust{%
591     \leftskip\the\leftskip
592     \rightskip\the\rightskip
593     \parfillskip\the\parfillskip\relax}%
594   \leftskip\z@\@plus -1fil%
595   \rightskip\z@\@plus 1fil%
596   \parfillskip\z@skip
597   \noindent\hskip\z@\@plus 2fil%
598   \@setpar{\@@par\@restorepar\caption@normaladjust}}

\centerlast This is based on code from Anne Brüggemann-Klein[?]
599 \providecommand\centerlast{%
600   \let\\@centercr
601   \leftskip\z@\@plus 1fil%

```

```

602 \rightskip\z@\@plus -1fil%
603 \parfillskip\z@\@plus 2fil\relax}

```

1.13.1 The ragged2e package

We also support the upper-case commands offered by the ragged2e package. Note that these just map to their lower-case variants if the ragged2e package is not available.

```

604 \DeclareCaptionJustification{Centering}{%
605   \caption@ragged\Centering\centering}
606 \DeclareCaptionJustification{RaggedLeft}{%
607   \caption@ragged\RaggedLeft\raggedleft}
608 \DeclareCaptionJustification{RaggedRight}{%
609   \caption@ragged\RaggedRight\raggedright}

```

\caption@ragged \caption@ragged will be basically defined as

```

\AtBeginDocument{\IfFileExists{ragged2e.sty}%
  {\RequirePackage{ragged2e}\let\caption@ragged\@firstoftwo}%
  {\let\caption@ragged\@secondoftwo}}

```

but with an additional warning if the ragged2e package is not loaded (yet). (This warning will be type out only one time per option, that's why we need the caption\string#1 stuff.) Furthermore we load the ragged2e package, if needed and available.

```

610 \newcommand*\caption@ragged{%
611   \caption@Debug{We need ragged2e}%
612   \protected@write\@auxout{}\string\caption@newlabel{ragged2e}{}%
613   \global\let\caption@ragged\caption@@ragged
614   \caption@ragged}

615 \caption@AtBeginDocument{%
616   \@ifundefined{caption@r@ragged2e}{%
617     \newcommand*\caption@@ragged{%
618       \caption@Warning{%
619         'ragged2e' support has been changed.\MessageBreak
620         Rerun to get captions right}%
621       \global\let\caption@ragged\@secondoftwo % suppress further warnings
622       \caption@ragged}%
623   }{%
624     \caption@Debug{We load ragged2e}%
625     \IfFileExists{ragged2e.sty}{%
626       \RequirePackage{ragged2e}%
627       \let\caption@@ragged\@firstoftwo
628     }{%
629       \newcommand*\caption@@ragged[2]{%
630         \@ifundefined{caption\string#1}{%
631           \caption@Warning{%
632             'ragged2e' package not loaded, therefore\MessageBreak
633             substituting \string#2 for \string#1\MessageBreak}%
634           \global\@namedef{caption\string#1}{}%
635           #2}%

```

```

636     }%
637   }}

```

1.14 Vertical spaces before and after captions

`\abovecaptionskip` `\belowcaptionskip` Usually these skips are defined within the document class, but some document classes don't do so.

```

638 \@ifundefined{abovecaptionskip}{%
639   \newlength\abovecaptionskip\setlength\abovecaptionskip{10\p@}}{}
640 \@ifundefined{belowcaptionskip}{%
641   \newlength\belowcaptionskip\setlength\belowcaptionskip{0\p@}}{}

642 \DeclareCaptionOption{aboveskip}{\setlength\abovecaptionskip{#1}}
643 \DeclareCaptionOption{belowskip}{\setlength\belowcaptionskip{#1}}
644 \DeclareCaptionOption{skip}{\setlength\abovecaptionskip{#1}}

```

1.15 Positioning

These macros handle the right position of the caption. Note that the position is actually *not* controlled by the `caption3` kernel options, but by the user (or a specific package like the `float` package) instead. The user can put the `\caption` command wherever he likes! So this stuff is only to give us a *hint* where to put the right skips, the user usually has to take care for himself that this hint actually matches the right position.

```

645 \DeclareCaptionOption{position}{\caption@setposition{#1}}

```

`\caption@setposition` `\caption@setposition{position}`

Selecting the caption position means that we put `\caption@position` to the right value. *Please do not use the internal macro `\caption@position` in your own package or document, but use the wrapper macro `\caption@iftop` instead.*

```

646 \newcommand*\caption@setposition[1]{%
647   \caption@ifinlist{#1}{d,default}{%
648     \let\caption@position\caption@defaultpos
649   }{\caption@ifinlist{#1}{t,top,above}{%
650     \let\caption@position\@firstoftwo
651   }{\caption@ifinlist{#1}{b,bottom,below}{%
652     \let\caption@position\@secondoftwo
653   }{\caption@ifinlist{#1}{a,auto}{%
654     \let\caption@position\@undefined
655   }{%
656     \caption@Error{Undefined position `#1'}%
657   }}}}%

```

`\caption@defaultpos` The default 'position' is 'auto', this means that the `caption` package will try to guess the current position of the caption. (But in many cases, for example in `longtables`, this is doomed to fail!)

The setting 'bottom' corresponds to the `\@makecaption` implementation in the standard $\text{\LaTeX}$ document classes, but 'auto' should give better results in most cases.

```

658 %\caption@setdefaultpos{a}% default = auto
659 \let\caption@defaultpos\@undefined

\caption@iftop \caption@iftop{\true-code}{\false-code}
(If the position= is set to auto we assume a bottom position here.)
660 \newcommand*\caption@iftop{%
661   \ifx\caption@position\@undefined
662     \let\caption@position\@secondoftwo
663 %   = \caption@setposition b%
664   \fi
665   \caption@position}

\caption@fixposition \caption@fixposition
This macro checks if the ‘position’ is set to ‘auto’. If yes, \caption@autoposition
will be called to set \caption@position to a proper value we can actually use.
666 \newcommand*\caption@fixposition{%
667   \ifx\caption@position\@undefined
668     \caption@autoposition
669   \fi}

\caption@autoposition \caption@autoposition
We guess the current position of the caption by checking \prevdepth.
A different solution would be setting the \spacefactor to something not much less
than 1000 (for example 994) in \caption@start and checking this value here by
\ifnum\spacefactor=994. (It’s implemented in the threeparttable package[?] this
way.)
Another idea would be checking \@ifminipage, but since some packages typeset the
caption within a simple \vbox this does not seem to be a good one.
670 \newcommand*\caption@autoposition{%
671   \ifvmode
672     \edef\caption@tempa{\the\prevdepth}%
673     \caption@Debug{\protect\prevdepth=\caption@tempa}%
674     \ifdim\prevdepth>-\p@
675       \let\caption@position\@secondoftwo
676     \else
677       \let\caption@position\@firstoftwo
678     \fi
679 %   = \caption@setposition{\ifdim\prevdepth>-\p@ b\else t\fi}%
680   \else
681     \caption@Debug{no \protect\prevdepth}%
682     \let\caption@position\@secondoftwo
683 %   = \caption@setposition b%
684   \fi}

\caption@setautoposition \caption@setautoposition{\position}
replaces the above algorithm by a different one (or a fixed position setting).
685 \newcommand*\caption@setautoposition[1]{%
686   \def\caption@autoposition{\caption@setposition{#1}}}
```

1.16 Hooks

```
\AtBeginCaption \AtBeginCaption {<code>}  
\AtEndCaption \AtEndCaption {<code>}
```

These hooks can be used analogous to `\AtBeginDocument` and `\AtEndDocument`.

```
687 \newcommand*\caption@beginhook{}  
688 \newcommand*\caption@endhook{}  
689 \newcommand*\AtBeginCaption{\l@addto@macro\caption@beginhook}  
690 \newcommand*\AtEndCaption{\l@addto@macro\caption@endhook}
```

1.17 Lists

```
691 \DeclareCaptionOption{listof}{\caption@set@bool\caption@iflof{#1}}
```

```
\DeclareCaptionListFormat \DeclareCaptionListFormat {<name>} {<code with #1 and #2>}
```

```
692 \newcommand*\DeclareCaptionListFormat[2]{%  
693   \global\@namedef{caption@lstfmt@#1}##1##2{#2}}  
694 \@onlypreamble\DeclareCaptionListFormat
```

```
695 \DeclareCaptionOption{listofformat}{\caption@setlistofformat{#1}}
```

```
\caption@setlistofformat \caption@setlistofformat {<name>}
```

Selecting a caption list-of format simply means saving the code (in `\caption@lstfmt`).

```
696 \newcommand*\caption@setlistofformat[1]{%  
697   \@ifundefined{caption@lstfmt@#1}%  
698     {\caption@Error{Undefined list-of format `#1'}}%  
699     {\expandafter\let\expandafter\caption@lstfmt  
700       \csname caption@lstfmt@#1\endcsname}}
```

There are five pre-defined list formats, taken from the `subfig` package.

```
701 \DeclareCaptionListFormat{empty}{}  
702 \DeclareCaptionListFormat{simple}{#1#2}  
703 \DeclareCaptionListFormat{parens}{#1 (#2)}  
704 \DeclareCaptionListFormat{subsimple}{#2}  
705 \DeclareCaptionListFormat{subparens}{ (#2)}  
706 \def\caption@lstfmt@default{\caption@lstfmt@subsimple}
```

1.18 Debug option

```
707 \DeclareCaptionOption{debug}[1]{%  
708   \caption@set@bool\caption@ifdebug{#1}%  
709   \let\caption@iftypewarning\caption@ifdebug  
710   \caption@ifdebug  
711     {\def\caption@Debug{\PackageInfo{caption}}}%  
712     {\let\caption@Debug\@gobble}}  
713 \DeclareOption{debug}{\setkeys{caption}{debug}}  
714 \setkeys{caption}{debug=0}
```

1.19 Document classes & Babel support

```
715 \caption@CheckCommand\@makecaption{\let\@makecaption\@undefined}
```

1.19.1 The standard L^AT_EX classes

```
716 \caption@CheckCommand\@makecaption{%
717 % article|report|book [2005/09/16 v1.4f Standard LaTeX document class]
718 \long\def\@makecaption#1#2{%
719   \vskip\abovecaptionskip
720   \sbox\@tempboxa{#1: #2}%
721   \ifdim \wd\@tempboxa >\hsize
722     #1: #2\par
723   \else
724     \global \@minipagefalse
725     \hb@xt@\hsize{\hfil\box\@tempboxa\hfil}%
726   \fi
727   \vskip\belowcaptionskip}}
```

1.19.2 The $\mathcal{A}\mathcal{M}\mathcal{S}$ & SMF classes

```
728 \@ifundefined{@captionheadfont}{}{%
729   \caption@CheckCommand\@makecaption{%
730     % amsart|amsproc|amsbook [2004/08/06 v2.20]
731     \long\def\@makecaption#1#2{%
732       \setbox\@tempboxa\vbox{\color@setgroup
733         \advance\hsize-2\captionindent\noindent
734         \@captionfont\@captionheadfont#1\@xp\ifnotempty\@xp
735           {\@cdr#2\@nil}{.\@captionfont\upshape\enspace#2}%
736         \unskip\kern-2\captionindent\par
737         \global\setbox\@ne\lastbox\color@endgroup}%
738       \ifhbox\@ne % the normal case
739         \setbox\@ne\hbox{\unhbox\@ne\unskip\unskip\unpenalty\unkern}%
740       \fi
741       \ifdim\wd\@tempboxa=\z@ % this means caption will fit on one line
742         \setbox\@ne\hbox to\columnwidth{\hss\kern-2\captionindent\box\@ne\hss}%
743       \else % tempboxa contained more than one line
744         \setbox\@ne\vbox{\unvbox\@tempboxa\parskip\z@skip
745           \noindent\unhbox\@ne\advance\hsize-2\captionindent\par}%
746       \fi
747       \ifnum\@tempcnta<64 % if the float IS a figure...
748         \addvspace\abovecaptionskip
749         \hbox to\hsize{\kern\captionindent\box\@ne\hss}%
750       \else % if the float IS NOT a figure...
751         \hbox to\hsize{\kern\captionindent\box\@ne\hss}%
752         \nobreak
753         \vskip\belowcaptionskip
754       \fi
755     \relax
756   }}
757 \caption@CheckCommand\@makecaption{%
```

```

758 % smfart|smfbook [1999/11/15 v1.2f Classe LaTeX pour les articles publies par L
759 \long\def\@makecaption#1#2{%
760   \ifdim\captionindent>.1\hsize \captionindent.1\hsize \fi
761   \setbox\@tempboxa\vbox{\color@setgroup
762     \advance\hsize-2\captionindent\noindent
763     \@captionfont\@captionheadfont#1\@xp\@ifnotempty\@xp
764       {\@cdr#2\@nil}\@addpunct{.}\@captionfont\upshape\enspace#2}%
765     \unskip\kern-2\captionindent\par
766     \global\setbox\@ne\lastbox\color@endgroup}%
767   \ifhbox\@ne % the normal case
768     \setbox\@ne\hbox{\unhbox\@ne\unskip\unskip\unpenalty\unkern}%
769   \fi
770   \ifdim\wd\@tempboxa=\z@ % this means caption will fit on one line
771     \setbox\@ne\hbox to\columnwidth{\hss\kern-2\captionindent\box\@ne\hss}%
772     \@tempdima\wd\@ne\advance\@tempdima-\captionindent
773     \wd\@ne\@tempdima
774   \else % tempboxa contained more than one line
775     \setbox\@ne\vbox{\rightskip=0pt plus\captionindent\relax
776       \unvbox\@tempboxa\parskip\z@skip
777       \noindent\unhbox\@ne\advance\hsize-2\captionindent\par}%
778   \fi
779   \ifnum\@tempcnta<64 % if the float IS a figure...
780     \addvspace\abovecaptionskip
781     \noindent\kern\captionindent\box\@ne
782   \else % if the float IS NOT a figure...
783     \noindent\kern\captionindent\box\@ne
784     \nobreak
785     \vskip\belowcaptionskip
786   \fi
787   \relax
788 }

789 \let\captionmargin\captionindent % set to 3pc by AMS class
790 \begingroup\edef\@tempa{\endgroup
791   \noexpand\caption@g@addto@list\noexpand\caption@sty@default
792     {margin=\the\captionmargin
793       \@ifundefined{smf@makecaption}{},{,maxmargin=.1\linewidth}}}}
794 \@tempa
795 \caption@g@addto@list\caption@sls@default{margin*=.5\captionmargin}
796 \DeclareCaptionLabelSeparator{default}{.\enspace}
797 \DeclareDefaultCaptionFont{font}{\@captionfont}
798 \DeclareDefaultCaptionFont{labelfont}{\@captionheadfont}
799 \DeclareDefaultCaptionFont{textfont}{\@captionfont\upshape}
800 \captionsetup[figure]{position=b}
801 \captionsetup[table]{position=t}
802 }

```

1.19.3 The beamer class

```

803 \@ifclassloaded{beamer}{%

```

```

804 \caption@CheckCommand\beamer@makecaption{%
805   % beamerbaselocalstructure.sty,v 1.53 2007/01/28 20:48:21 tantau
806   \long\def\beamer@makecaption#1#2{%
807     \def\insertcaptionname{\csname#1name\endcsname}%
808     \def\insertcaptionnumber{\csname the#1\endcsname}%
809     \def\insertcaption{#2}%
810     \nobreak\vskip\abovecaptionskip\nobreak
811     \sbox\@tempboxa{\usebeamertemplate**{caption}}%
812     \ifdim \wd\@tempboxa >\hsize
813       \usebeamertemplate**{caption}\par
814     \else
815       \global \@minipagefalse
816       \hb@xt@\hsize{\hfil\box\@tempboxa\hfil}%
817     \fi
818     \nobreak\vskip\belowcaptionskip\nobreak}}
819 \DeclareCaptionLabelFormat{default}{#1}
820 \DeclareDefaultCaptionJustification{\raggedright}
821 \DeclareDefaultCaptionFont{font}{%
822   \usebeamerfont*{caption}%
823   \usebeamercolor[fg]{caption}}
824 \DeclareDefaultCaptionFont{labelfont}{%
825   \usebeamercolor[fg]{caption name}%
826   \usebeamerfont*{caption name}}

```

If the beamer document class is used, we offer a beamer template called ‘caption3’ which can be used with option ‘beamer’ or `\setbeamertemplate{caption}[caption3]`. (Note that this is of no use when the caption package is used, too.)

```

827 \defbeamertemplate{caption}{caption3}{%
828   \caption@make\insertcaptionname\insertcaptionnumber\insertcaption}
829 \DeclareOption{beamer}{%
830   % \usebeamertemplate**{caption} will set font
831   \DeclareDefaultCaptionFont{font}{}%
832   \setbeamertemplate{caption}[caption3]}
833 {}

```

1.19.4 The KOMA-Script classes

```

834 \ifundefined{scr@caption}{}{%
835   \caption@CheckCommand\@makecaption{%
836     % scrartcl|scrreprt|scrbook [2007/03/07 v2.97a KOMA-Script document class]
837     \long\def\@makecaption#1#2{%
838       \if@captionabove
839         \vskip\belowcaptionskip
840       \else
841         \vskip\abovecaptionskip
842       \fi
843       \@@makecaption\@firstofone{#1}{#2}%
844       \if@captionabove
845         \vskip\abovecaptionskip

```

```

846     \else
847         \vskip\belowcaptionskip
848     \fi}}
849 \DeclareCaptionFormat{default} [#1#2#3\par] {%
850     \ifdofullc@p
851         \caption@ifin@list\caption@lsep@crlist\caption@lsepname
852         {\caption@Error{%
853             The option 'labelsep=\caption@lsepname' does not work\MessageBreak
854             with \noexpand\setcaphanging (which is set by default)}}%
855         {\caption@fmt@hang{#1}{#2}{#3}}}%
856     \else
857         #1#2%
858         \ifdim\cap@indent<\z@
859             \par
860             \noindent\hspace*{-\cap@indent}%
861         \else\if@capbreak
862             \par
863         \fi\fi
864         #3\par
865     \fi}
866 \DeclareCaptionLabelSeparator{default}{\captionformat}
867 \DeclareDefaultCaptionFont{font}{\scr@fnt@caption}
868 \DeclareDefaultCaptionFont{labelfont}{\scr@fnt@captionlabel}
869 }

```

1.19.5 The NTG Dutch classes

```

870 \@ifundefined{CaptionFonts}{}{%
871     \caption@CheckCommand\@makecaption{%
872         % artikel|rapport|boek [2004/06/07 v2.1a NTG LaTeX document class]
873         \long\def\@makecaption#1#2{%
874             \vskip\abovecaptionskip
875             \sbox\@tempboxa{\CaptionLabelFont#1:} \CaptionTextFont#2}%
876             \ifdim \wd\@tempboxa >\hsize
877                 {\CaptionLabelFont#1:} \CaptionTextFont#2\par
878             \else
879                 \global \@minipagefalse
880                 \hb@xt@\hsize{\hfil\box\@tempboxa\hfil}%
881             \fi
882             \vskip\belowcaptionskip}}
883 \DeclareDefaultCaptionFont{labelfont}{\CaptionLabelFont}
884 \DeclareDefaultCaptionFont{textfont}{\CaptionTextFont}
885 }

```

1.19.6 The frenchb Babel option

```

886 \@ifundefined{FB@makecaption}{}{%
887     \caption@CheckCommand\@makecaption{%
888         % frenchb.ldf [2005/02/06 v1.6g French support from the babel system]

```

```

889 \long\def\@makecaption#1#2{%
890 \vskip\abovecaptionskip
891 \sbox\@tempboxa{#1\CaptionSeparator #2}%
892 \ifdim \wd\@tempboxa >\hsize
893 #1\CaptionSeparator #2\par
894 \else
895 \global \@minipagefalse
896 \hb@xt@\hsize{\hfil\box\@tempboxa\hfil}%
897 \fi
898 \vskip\belowcaptionskip}}
899 \ifx\@makecaption\STD@makecaption
900 \DeclareCaptionLabelSeparator{default}{\CaptionSeparator}
901 \def\caption@frenchb{% supress frenchb warning
902 \let\STD@makecaption\@makecaption
903 \let\FB@makecaption\@makecaption}
904 \else
905 \ifx\@makecaption\@undefined\else
906 \PackageInfo{caption}{%
907 The definition of \protect\@makecaption\space
908 has been changed,\MessageBreak
909 frenchb will NOT customize it}%
910 \fi
911 \fi
912 }

```

1.19.7 The frenchle/pro package

```

913 \@ifundefined{frenchTeXmods}{}{%
914 \caption@CheckCommand\@makecaption{%
915 % french(1e).sty [2006/10/03 The french(1e) package /V5,9991/]
916 % french(1e).sty [2007/06/28 The french(1e) package /V5,9994/]
917 \def\@makecaption#1#2{%
918 \ifFTY%
919 \def\@secondofmany##1##2\void{##2}%
920 \def\@tempa{\@secondofmany#2\void}%
921 \ifx\@tempa\empty%
922 \let\captionseparator\empty%
923 \fi%
924 \@mcORI{#1}{\relax\captionfont{#2}}%
925 \else
926 \@mcORI{#1}{#2}%
927 \fi}}
928 \caption@CheckCommand\@makecaption{%
929 % french(1e).sty [2007/02/11 The french(1e) package /V5,9993/]
930 \def\@makecaption#1#2{%
931 \ifFTY%
932 \def\@secondofmany##1##2\void{##2}%
933 \protected@edef\@tempa{\@secondofmany#2\void}%
934 \ifx\@tempa\empty%

```

```

935         \let\captionseparator\empty%
936     \fi%
937     \mcORI{#1}{\relax\captionfont{#2}}%
938 \else
939     \mcORI{#1}{#2}%
940 \fi}}
941 \DeclareDefaultCaptionFont{textfont}{\itshape}%
942 \DeclareCaptionLabelSeparator{default}{\captionseparator\space}%
943 }

```

1.20 Execution of options

```

944 \captionsetup{style=default,position=default,listof=1,listofformat=default,%
945             twoside=\if@twoside 1\else 0\fi}
946 \ProcessOptions*
947 \caption@ifcheckcommand{%
948   \caption@setbool{documentclass}{1}%
949 }{%
950   \caption@setbool{documentclass}{0}%
951   \PackageInfo{caption}{%
952     Unknown document class (or package),\MessageBreak
953     standard defaults will be used}%
954   \caption@debug{\string\@makecaption\space=\space\meaning\@makecaption\@gobble}%
955 }

```

1.21 Making an ‘List of’ entry

```

\caption@addcontentsline \caption@addcontentsline{<type>}{<list entry>}

```

Makes an entry in the list-of-whatever, if requested, i.e. the argument *<list entry>* is not empty and `listof=` was set to true.

```

956 \newcommand*\caption@addcontentsline[2]{%
957   \caption@iflof%
958   {\def\@tempa{#2}}%
959   {\let\@tempa\@empty}%
960   \ifx\@tempa\@empty \else
961     {\let\\space
962       \addcontentsline{\csname ext@#1\endcsname}{#1}%
963                       {\protect\numberline
964                        {\caption@lstfmt{\@nameuse{p@#1}}{\@nameuse{the#1}}}%
965                        {\ignorespaces #2}}}%
966   \fi}

```

1.22 Typesetting the caption

```

\ifcaption@star If the starred form of \caption is used, this will be set to true. (It will be reset to
false at the end of \caption@@make.)
967 \newif\ifcaption@star

```

```

\caption@fnum \caption@fnum{<float type>}
Typesets the caption label; as replacement for \fnum@<float type>.
968 \newcommand*\caption@fnum[1]{%
969   \caption@lfmt{\@nameuse{#1name}}{\@nameuse{the#1}}{\theContinuedFloat}}

\caption@make \caption@make{<float name>}{<ref. number>}{<text>}
Typesets the caption.
970 \newcommand\caption@make[2]{%
971   \caption@@make{\caption@lfmt{#1}{#2}}{\theContinuedFloat}}

\caption@@make \caption@@make{<caption label>}{<caption text>}
972 \newcommand\caption@@make[2]{%
973   \begingroup
974   \caption@stepcounter
975   \caption@beginhook

Check margin, if \caption@minmargin or \caption@maxmargin is set
976   \ifx\caption@maxmargin\undefined \else
977     \ifdim\captionmargin>\caption@maxmargin\relax
978       \captionmargin\caption@maxmargin\relax
979     \fi
980   \fi
981   \ifx\caption@minmargin\undefined \else
982     \ifdim\captionmargin<\caption@minmargin\relax
983       \captionmargin\caption@minmargin\relax
984     \fi
985   \fi

Special single-line treatment (option singlelinecheck=)
986   \caption@ifslc{\caption@slc{#1}{#2}\captionwidth\relax}{}%

Typeset the left margin (option margin=)
987   \caption@calcmargin
988   \@tempdima\captionmargin
989   \ifdim\captionmargin@=\z@ \else
990     \caption@ifoddpage{}{\advance\@tempdima\captionmargin}%
991   \fi
992   \caption@ifh{\advance\@tempdima\caption@indent}%
993   \hspace\@tempdima

We actually use a \vbox of width \captionwidth - \caption@indent to
typeset the caption.
Note: \captionindent is not supported if the caption format was defined with \DeclareCaptionFormat*.
994   \@tempdima\captionwidth
995   \caption@ifh{\advance\@tempdima-\caption@indent}%
996   \captionbox\@tempdima{%

Typeset the indentation (option indention=)
Bugfix 04-05-05: \hskip-\caption@indent replaced by \ifdim\caption@indent=\z@...
997   \caption@ifh{%
998     \ifdim\caption@indent=\z@

```

```

999      \leavevmode
1000     \else
1001       \hskip-\caption@indent
1002     \fi}%

Typeset the caption itself and close the \captionbox
1003     \caption@@@make{#1}{#2}}%

Typeset the right margin (option margin=)
1004     \@tempdima\captionmargin
1005     \ifdim\captionmargin@=\z@ \else
1006       \caption@ifoddpage{\advance\@tempdima\captionmargin@}{}%
1007     \fi
1008     \hspace\@tempdima
1009     \caption@endhook
1010     \endgroup
1011     \global\caption@starfalse}

\caption@calcmargin \caption@calcmargin
Calculate \captionmargin & \captionwidth, so both contain valid values.
1012 \newcommand*\caption@calcmargin{%
1013   \ifdim\captionwidth=\z@
1014     \captionwidth\linewidth
1015     \advance\captionwidth by -2\captionmargin
1016     \advance\captionwidth by -\captionmargin@
1017   \else
1018     \captionmargin\linewidth
1019     \advance\captionmargin by -\captionwidth
1020     \divide\captionmargin by 2
1021     \captionmargin@\z@
1022   \fi

1023   \caption@Debug{%
1024     \string\hsize=\the\hsize,
1025     \string\linewidth=\the\linewidth,\MessageBreak
1026     \string\leftmargin=\the\leftmargin,
1027     \string\rightmargin=\the\rightmargin,\MessageBreak
1028     \string\margin=\the\captionmargin,
1029     \string\margin@=\the\captionmargin@,
1030     \string\width=\the\captionwidth}%
1031 }

\caption@slc \caption@slc{\label}{\text}{\width}{\extra code}
This one does the single-line-check.
1032 \newcommand\caption@slc[4]{%
1033   \caption@Debug{Begin SLC}%
1034   \begingroup
1035   \caption@singleline
1036   \let\caption@hj\@empty

```

```

1037 \caption@calcmargin % calculate #3 if necessary
1038 \caption@prepareslc
1039 \sbox\@tempboxa{\caption@@@make{#1}{#2}}%
1040 \ifdim\wd\@tempboxa>#3%
1041   \endgroup
1042 \else
1043   \endgroup
1044   \caption@singleline
1045   #4%
1046 \fi
1047 \caption@Debug{End SLC}}

1048 \newcommand*\caption@singleline{%
1049   \expandafter\caption@setup\expandafter{\caption@opt@singleline}%
1050   \let\caption@fmt\caption@slfmt}

\caption@prepareslc \caption@prepareslc
  Re-define anything which would disturb the single-line-check.
1051 \newcommand*\caption@prepareslc{%
1052   \let\@footnotetext\@gobble\let\@endnotetext\@gobble
1053   \def\label{\caption@withoptargs\@gobbletwo}%
1054   \let\stepcounter\caption@l@stepcounter
1055   \let\refstepcounter\stepcounter\let\H\refstepcounter\stepcounter}

1056 \newcommand*\caption@l@stepcounter[1]{\advance\c@#1\endcsname\@ne\relax}

\captionbox \captionbox{\langle width \rangle}{\langle contents \rangle}
  This macro defines the box which surrounds the caption paragraph.
1057 \newcommand*\captionbox{\parbox[b]}

\caption@@@make \caption@@@make{\langle caption label \rangle}{\langle caption text \rangle}
  This one finally typesets the caption paragraph, without margin and indentation.
1058 \newcommand\caption@@@make[2]{%
  If the label is empty, we use no caption label separator.
1059   \sbox\@tempboxa{#1}%
1060   \ifdim\wd\@tempboxa=\z@
1061     \let\caption@lsep\relax
1062 %   \@capbreakfalse
1063   \fi

  If the text is empty, we use no caption label separator, too.
1064   \caption@ifempty{#2}{%
1065     \let\caption@lsep\relax
1066 %   \@capbreakfalse
1067     \let\caption@ifstrut\@secondoftwo
1068   }%

  Take care that \caption@parindent and \caption@hangindent will be used
  to typeset the paragraph.
1069   \@setpar{\@par\caption@par}\caption@@par

```

Finally typeset the caption.

```

1070 \caption@hj\captionsize\captionfont\caption@fmt
1071     {\ifcaption@star\else{\captionlabelfont#1}\fi}%
1072     {\ifcaption@star\else{\caption@iflf\captionlabelfont\caption@lsep}\fi}%
1073     {\captiontextfont
1074       \caption@ifstrut{\vrule\@height\ht\strutbox\@width\z@}{}}%
1075       \nobreak\hskip\z@skip % enable hyphenation
1076       \caption@tfmt{#2}%
1077 %       \caption@ifstrut{\vrule\@height\z@\@depth\dp\strutbox\@width\z@}{}}%
1078       \caption@ifstrut{\ifhmode\@finalstrut\strutbox\fi}{}}%
1079     \par}}

```

`\caption@ifempty` `\caption@ifempty{<text>}{<true>}{no <false>}`

This one tests if the *<text>* is actually empty.

Note: This will be done without expanding the text, therefore this is far away from being bullet-proof.

Note: This macro is re-defining itself so only the first test (in a group) will actually be done.

```

1080 \newcommand\caption@ifempty[1]{%
1081   \def\caption@tempa{#1}%
1082   \def\caption@tempb{\ignorespaces}%
1083   \ifx\caption@tempa\caption@tempb
1084     \let\caption@tempa\@empty
1085   \fi
1086   \ifx\caption@tempa\@empty
1087     \let\caption@ifempty\@secondoftwo
1088   \else
1089     \let\caption@ifempty\@gobbletwo
1090   \fi
1091   \caption@ifempty\@unused}

```

`\caption@@par` `\caption@@par`

This command will be executed with every `\par` inside the caption.

```

1092 \newcommand*\caption@@par{%
1093   \parindent\caption@parindent\hangindent\caption@hangindent}%

```

1.23 Bonus features

```

\DeclareFloatingEnvironment \DeclareFloatingEnvironment[<file ext>]{<environment>}[<name>][<list name>]
1094 \providecommand*\DeclareFloatingEnvironment{%
1095   \@ifnextchar[%]
1096     \caption@@DeclareFloatingEnvironment\caption@DeclareFloatingEnvironment}
1097 \@onlypreamble\DeclareFloatingEnvironment

1098 \newcommand*\caption@DeclareFloatingEnvironment[1]{%
1099   \caption@@DeclareFloatingEnvironment[lo#1]{#1}}
1100 \@onlypreamble\caption@DeclareFloatingEnvironment

1101 \def\caption@@DeclareFloatingEnvironment[#1]#2{%
1102   \@ifundefined{c@chapter}%

```

```

1103     {\newcounter{#2}%
1104      \@namedef{the#2}{\arabic{#2}}%
1105      \@namedef{theH#2}{\arabic{#2}}}%
1106     {\newcounter{#2}[chapter]%
1107      \@namedef{the#2}{\ifnum\c@chapter>\z@ \thechapter.\fi \arabic{#2}}%
1108      \@namedef{theH#2}{\theHchapter.\arabic{#2}}}%
1109     \@namedef{fps@#2}{tbp}%
1110     \@ifundefined{c@float@type}%
1111     {\newcounter{float@type}%
1112      \setcounter{float@type}{\@ifundefined{c@figure}14}}%
1113     {}%
1114     \caption@Debug{float type `#2'=\the\value{float@type}}%
1115     \expandafter\xdef\csname ftype@#2\endcsname{\the\value{float@type}}%
1116     \addtocounter{float@type}{\value{float@type}}%
1117     \@namedef{ext@#2}{#1}%
1118     \@namedef{fnum@#2}{\@nameuse{#2name}\nobreakspace\@nameuse{the#2}}%
1119     \newenvironment{#2}{\@float{#2}}{\end@float}%
1120     \newenvironment{#2*}{\@dblfloat{#2}}{\end@dblfloat}%
1121     \expandafter\newcommand\csname listof#2s\endcsname{\begin@group
1122       \expandafter\let\expandafter\listfigurename\csname list#2name\endcsname
1123       \expandafter\let\expandafter\ext@figure\csname ext@#2\endcsname
1124       \let\caption@ORI@starttoc\@starttoc
1125       \renewcommand*\@starttoc[1]{%
1126         \expandafter\caption@ORI@starttoc\expandafter{\ext@figure}}%
1127       \listoffigures
1128       \endgroup}%
1129     \@ifundefined{l@figure}%
1130     {\@namedef{l@#2}{\@dottedtocline{1}{1.5em}{2.3em}}}%
1131     {\expandafter\let\csname l@#2\endcsname{l@figure}}%
1132     \expandafter\newcommand\csname #2name\endcsname{}%
1133     \edef\@tempa{%
1134       \def\noexpand\@tempa{\expandafter\noexpand\@car#2\@nil}}%
1135     \uppercase\expandafter{\@tempa}%
1136     \edef\@tempb{%
1137       \noexpand\g@addto@macro\noexpand\@tempa{\expandafter\@cdr#2\@nil}}%
1138     \@tempb
1139     \expandafter\let\csname #2name\endcsname\@tempa
1140     \expandafter\newcommand\csname list#2name\endcsname{}%
1141     \expandafter\xdef\csname list#2name\endcsname{List of \@tempa s}%
1142     \@cons\caption@floatlist{{#2}}%
1143     \@ifnextchar[{\caption@@@DeclareFloatingEnvironment{#2}}\relax}
1144 \@onlypreamble\caption@@@DeclareFloatingEnvironment

1145 \def\caption@@@DeclareFloatingEnvironment#1[#2]{%
1146   \@namedef{#1name}{#2}%
1147   \@ifnextchar[{\caption@@@DeclareFloatingEnvironment{#1}}\relax}
1148 \@onlypreamble\caption@@@DeclareFloatingEnvironment

1149 \def\caption@@@DeclareFloatingEnvironment#1[#2]{%
1150   \@namedef{list#1name}{#2}}
1151 \@onlypreamble\caption@@@DeclareFloatingEnvironment

```

```

1152 \newcommand*\caption@floatlist{}

\ForFloatingEnvironments \ForFloatingEnvironments{<code with #1>}
1153 \providecommand\ForFloatingEnvironments[1]{%
1154   \caption@AtBeginDocument{%
1155     \def\@elt##1{#1}%
1156     \caption@floatlist
1157     \let\@elt\relax}}%
1158 \@onlypreamble\ForFloatingEnvironments

```

1.24 subfig package adoptions

Unfortunately the author of the subfig package does not seem to maintain his package anymore, so we have to make several adoptions here.

```

1159 \caption@AtBeginDocument{%
1160   \def\@tempa{\@ifstar\sf@@subref\sf@subref}%
1161   \ifx\subref\@tempa
1162     \PackageInfo{caption3}{subfig package 1.2 or 1.3 is loaded\@gobble}%
1163     \let\caption@setfloattype\@gobble
1164 %   \let\sf@ifinlist\caption@ifinlist
1165     \let\sf@ifpositiontop\caption@iftop
1166     \def\@dottedxxxline#1#2#3#4#5#6#7{%
1167       \begingroup
1168         \caption@setfloattype{#1}%
1169         \caption@setoptions{subfloat}%
1170         \caption@setoptions{sub#1}%
1171         \ifnum #3>\@nameuse{c@#2depth}\else
1172           \@dottedtocline{\z@}{#4}{#5}{#6}{#7}%
1173         \fi
1174       \endgroup}%
1175     \let\sf@subfloat\sf@NEW@subfloat
1176     \caption@removefromoptlist{subfloat}%
1177     \caption@removefromoptlist{subfigure}%
1178     \caption@removefromoptlist{subtable}%
1179   \fi
1180   \let\sf@NEW@subfloat\@undefined}
1181 \def\sf@NEW@subfloat{%
1182   \begingroup
1183     \caption@setfloattype\@capttype
1184     \sf@ifpositiontop{%
1185       \maincaptiontoptrue
1186     }{%
1187       \maincaptiontopfalse
1188     }%
1189     \caption@setoptions{subfloat}%
1190     \caption@setoptions{sub\@capttype}%

```

```

1191 \let\sf@oldlabel=\label
1192 \let\label=\subfloat@label
1193 \ifmaincaptiontop\else
1194   \advance\@nameuse{c@\@capttype}\@ne
1195 \fi
1196 \refstepcounter{sub\@capttype}%
1197 \setcounter{sub\@capttype @save}{\value{sub\@capttype}}%
1198 \@ifnextchar [% %] match left bracket
1199   {\sf@@subfloat}%
1200   {\sf@@subfloat[\@empty]}}%

```

2 Main package

2.1 Identification

```
1201 \NeedsTeXFormat{LaTeX2e}[1994/12/01]
1202 \ProvidesPackage{caption}[2007/11/04 v3.1e Customising captions (AR)]
1203 %\@ifundefined{PackageRedefines}{}{\PackageRedefines{caption}{caption}}

\caption@Info Note: The \@gobble at the end of the 2nd argument of \PackageInfo suppresses the line
              number info. See TLC2[?], A.4.7, p885 for details.

1204 \newcommand*\caption@Info[1]{\PackageInfo{caption}{#1\@gobble}}
1205 \@onlypreamble\caption@Info
```

2.2 Loading the kernel

```
1206 \RequirePackage{caption3}[2007/11/04] % needs v3.1e or newer
```

2.3 Check against incompatible packages

```
1207 \@ifpackageloaded{caption2}{%
1208   \caption@Error{%
1209     You can't use both, the (obsolete) caption2 *and*\MessageBreak
1210     the (current) caption package}%
1211   \endinput
1212 }{}

1213 \caption@AtBeginDocument{%
1214   \@ifpackageloaded{ftcap}{%
1215     \caption@IncompatiblePackage{ftcap}{}%
1216   }{}%
1217   \@ifpackageloaded{nonfloat}{%
1218     \caption@IncompatiblePackage{nonfloat}{.\MessageBreak
1219       Furthermore you could use \noexpand\captionof instead of \MessageBreak
1220       \noexpand\figcaption or \string\tabcaption}%
1221   }{}%
1222   \@ifpackageloaded{topcapt}{%
1223     \caption@IncompatiblePackage{topcapt}{}%
1224   }{}%
1225 }

\caption@Incompatible Issue an 'incompatible' warning.

1226 \newcommand*\caption@IncompatiblePackage[2]{%
1227   \PackageWarningNoLine{caption}{%
1228     Usage of the #1 package together with the \MessageBreak
1229     caption package is strongly *not* recommended.\MessageBreak
1230     Instead of loading the #1 package you should \MessageBreak
1231     use the caption package option \MessageBreak
1232     \space\space 'tableposition=top' #2}}
1233 \@onlypreamble\caption@IncompatiblePackage
```

2.4 Check document class

```

1234 \caption@ifbool{documentclass}{\}%
1235 \PackageWarningNoLine{caption}{\%
1236   Unsupported document class (or package) detected,\MessageBreak
1237   usage of the caption package is *not* recommended}%
1238 \caption@Info{\string\@makecaption\space=\space\meaning\@makecaption}%
1239 }

```

2.5 Adaption to the $\mathcal{A}\mathcal{M}\mathcal{S}$ & SMF document classes

```

1240 \@ifundefined{@captionheadfont}{\}%
1241 \caption@Info{AMS or SMF document class}%
1242 \setlength\belowcaptionskip{0pt}% set to 12pt by AMS class
1243 }

```

2.6 Emulation of the KOMA-Script commands

```

1244 \@ifundefined{scr@caption}{\}%
1245 \caption@Info{KOMA-Script document class}%

```

Here we emulate the caption related commands and take over the caption related settings from the KOMA-Script classes.

```

\@tablecaptionabovetrue
\@tablecaptionabovefalse 1246 \def\@tablecaptionabovetrue{\captionsetup*[table]{position=t}}
1247 \def\@tablecaptionabovefalse{\captionsetup*[table]{position=b}}

1248 \if@tablecaptionabove
1249   \caption@AtBeginDocument{%
1250     \def\caption@tempa#1{%
1251       \ifpackageloaded{#1}{\caption@Error{%
1252         You can't use the global option\MessageBreak
1253         \space\space'tablecaptionabove'\MessageBreak
1254         and use the #1 package, too}}{\}%
1255       \caption@tempa{ftcap}%
1256       \caption@tempa{nonfloat}%
1257       \caption@tempa{topcapt}}%
1258   \@tablecaptionabovetrue
1259 \else
1260   \@tablecaptionabovefalse
1261 \fi

\onelinecaptionstrue
\onelinecaptionfalse 1262 \def\onelinecaptionstrue{\let\caption@ifslc\@firstoftwo}
1263 \def\onelinecaptionfalse{\let\caption@ifslc\@secondoftwo}

1264 \ifonelinecaptions
1265   \onelinecaptionstrue
1266 \else
1267   \onelinecaptionfalse
1268 \fi

```

```

\captionabovetrue
\captionabovefalse 1269 \def\captionabovetrue{\let\caption@position\@firstoftwo}
1270 \def\captionabovefalse{\let\caption@position\@secondoftwo}

\setcapindent
1271 \let\caption@KOMA@setcapindent\@setcapindent
1272 \renewcommand*\@setcapindent[1]{%
1273   \caption@KOMA@setcapindent{#1}\caption@setcapindent}

1274 \let\caption@KOMA@@setcapindent\@@setcapindent
1275 \renewcommand*\@@setcapindent[1]{%
1276   \caption@KOMA@@setcapindent{#1}\caption@setcapindent}

1277 \newcommand*\caption@setcapindent{%
1278   \captionsetup{indent=\ifdim\cap@indent<\z@\z@\else\cap@indent\fi}}

1279 \@ifundefined{cap@indent}{}{\caption@setcapindent}

\setcapwidth Note: The optional argument of \setcapwidth if not supported (yet), so we issue a warning if
used. (Since this does not seem to have an negative effect when used by the captionbeside
environment, we suppress the warning here.)
1280 \expandafter\let\expandafter\caption@KOMA@setcapwidth
1281   \csname\string\setcapwidth\endcsname
1282 \@namedef{\string\setcapwidth}[#1]#2{%
1283   \caption@KOMA@setcapwidth[#1]{#2}\caption@setcapwidth{#1}}

1284 \newcommand*\caption@setcapwidth[1]{%
1285   \ifx\#1\\\else
1286     \@ifundefined{cap@margin}{}{%
1287       \def\@tempa{captionbeside}%
1288       \ifx\@tempa\@currenvir\else\caption@Warning{%
1289         Ignoring optional argument [1] of \string\setcapwidth\MessageBreak}%
1290       \fi}%
1291   \fi
1292   \captionsetup{width=\cap@width}}

1293 \def\caption@tempa{\hsize}%
1294 \ifx\caption@tempa\cap@width \else
1295   \caption@setcapwidth{?}
1296 \fi

\setcapmargin
1297 \expandafter\let\expandafter\caption@KOMA@setcapmargin
1298   \csname\string\@setcapmargin\endcsname
1299 \@namedef{\string\@setcapmargin}[#1]#2{%
1300   \caption@KOMA@setcapmargin[#1]{#2}\caption@setcapmargin}

1301 \expandafter\let\expandafter\caption@KOMA@@setcapmargin
1302   \csname\string\@@setcapmargin\endcsname
1303 \@namedef{\string\@@setcapmargin}[#1]#2{%
1304   \caption@KOMA@@setcapmargin[#1]{#2}\caption@setcapmargin}

```

```

1305 \newcommand*\caption@setcapmargin{%
1306   \begingroup
1307     \let\onelinecaptionsfalse\relax
1308     \def\@twoside{0}%
1309     \def\if@twoside{\def\@twoside{1}\iffalse}%
1310     \cap@margin
1311     \def\@tempa{\endgroup}%
1312     \ifx\cap@left\hfill\else\ifx\cap@right\hfill\else
1313       \def\hspace##1##{\@firstofone}%
1314       \edef\@tempa{\endgroup
1315         \noexpand\captionsetup{%
1316           twoside=\@twoside,singlelinecheck=0,%
1317           margin={\cap@left,\cap@right}}}%
1318     \fi\fi
1319     \@tempa}

1320 \ifx\cap@margin\relax \else
1321   \caption@setcapmargin
1322 \fi

1323 }

```

2.7 Option for configuration files

```

1324 \DeclareCaptionOption*{config}[caption]{%
1325   \InputIfFileExists{#1.cfg}%
1326   {\typeout{*** Local configuration file #1.cfg used ***}}%
1327   {\caption@Warning{Configuration file #1.cfg not found}}}

```

2.8 Options for figure and table

```

1328 \DeclareCaptionOption{figureposition}{%
1329   \captionsetup*[figure]{position=#1}}
1330 \@onlypreamble@key{caption}{figureposition}

1331 \DeclareCaptionOption{tableposition}{%
1332   \caption@AtBeginDocument{%
1333     \def\caption@tempa##1{%
1334       \ifpackageloaded{##1}{\caption@Error{%
1335         You can't use the caption package option\MessageBreak
1336         \space\space'tableposition=#1'\MessageBreak
1337         and use the ##1 package, too}}{}}%
1338       \caption@tempa{ftcap}%
1339       \caption@tempa{nonfloat}%
1340       \caption@tempa{topcapt}}%
1341     \captionsetup*[table]{position=#1}}
1342 \@onlypreamble@key{caption}{tableposition}

1343 \DeclareCaptionOption{figurename}{\caption@SetFloatName{figure}{#1}}
1344 \@onlypreamble@key{caption}{figurename}
1345 \DeclareCaptionOption{tablename}{\caption@SetFloatName{table}{#1}}
1346 \@onlypreamble@key{caption}{tablename}

```

```

\caption@SetFloatName \caption@SetFloatName{<cmd>}{<value>}
1347 \newcommand*\caption@SetFloatName[2]{%
1348 \begingroup
1349 \def\@tempa{\endgroup\@firstofone}%
1350 \@ifundefined{language}{\@firstofone}%
1351 \@ifundefined{captions\language}{\@firstofone}%
1352 \def\@tempa{\endgroup
1353 \expandafter\addto\csname captions\language\endcsname}}%
1354 \@tempa{\@namedef{#1name}{#2}}%
1355 \@onlypreamble\caption@SetFloatName

```

2.9 caption v1.x compatibility options

```

1356 \DeclareCaptionOption{compatibility}[1]{\caption@setbool{compatibility}{#1}}
1357 \@onlypreamble@key{caption}{compatibility}

1358 \DeclareCaptionOptionNoValue*{normal}{%
1359 \caption@setformat{plain}%
1360 \caption@setjustification{justified}}
1361 \DeclareCaptionOptionNoValue*{isu}{%
1362 \caption@setformat{hang}%
1363 \caption@setjustification{justified}}
1364 \DeclareCaptionOptionNoValue*{hang}{%
1365 \caption@setformat{hang}%
1366 \caption@setjustification{justified}}
1367 \DeclareCaptionOptionNoValue*{center}{%
1368 \caption@setformat{plain}%
1369 \caption@setjustification{centering}}
1370 \DeclareCaptionOptionNoValue*{anne}{%
1371 \caption@setformat{plain}%
1372 \caption@setjustification{centerlast}}
1373 \DeclareCaptionOptionNoValue*{centerlast}{%
1374 \caption@setformat{plain}%
1375 \caption@setjustification{centerlast}}

1376 \DeclareCaptionOptionNoValue*{scriptsize}{\def\captionfont{\scriptsize}}
1377 \DeclareCaptionOptionNoValue*{footnotesize}{\def\captionfont{\footnotesize}}
1378 \DeclareCaptionOptionNoValue*{small}{\def\captionfont{\small}}
1379 \DeclareCaptionOptionNoValue*{normalsize}{\def\captionfont{\normalsize}}
1380 \DeclareCaptionOptionNoValue*{large}{\def\captionfont{\large}}
1381 \DeclareCaptionOptionNoValue*{Large}{\def\captionfont{\Large}}

1382 \DeclareCaptionOptionNoValue*{up}{\l@addto@macro\captionlabelfont\upshape}
1383 \DeclareCaptionOptionNoValue*{it}{\l@addto@macro\captionlabelfont\itshape}
1384 \DeclareCaptionOptionNoValue*{sl}{\l@addto@macro\captionlabelfont\slshape}
1385 \DeclareCaptionOptionNoValue*{sc}{\l@addto@macro\captionlabelfont\scshape}
1386 \DeclareCaptionOptionNoValue*{md}{\l@addto@macro\captionlabelfont\mdseries}
1387 \DeclareCaptionOptionNoValue*{bf}{\l@addto@macro\captionlabelfont\bfseries}
1388 \DeclareCaptionOptionNoValue*{rm}{\l@addto@macro\captionlabelfont\rmfamily}
1389 \DeclareCaptionOptionNoValue*{sf}{\l@addto@macro\captionlabelfont\sffamily}
1390 \DeclareCaptionOptionNoValue*{tt}{\l@addto@macro\captionlabelfont\ttfamily}

```

```

1391 \DeclareCaptionOptionNoValue*{nooneline}{\caption@setbool{slc}{0}}
1392 \caption@setbool{ruled}{0}
1393 \DeclareCaptionOptionNoValue*{ruled}{\caption@setbool{ruled}{1}}

```

2.10 caption2 v2.x compatibility options

```

1394 \DeclareCaptionOptionNoValue*{flushleft}{%
1395   \caption@setformat{plain}%
1396   \caption@setjustification{raggedright}}
1397 \DeclareCaptionOptionNoValue*{flushright}{%
1398   \caption@setformat{plain}%
1399   \caption@setjustification{raggedleft}}
1400 \DeclareCaptionOptionNoValue*{oneline}{\caption@setbool{slc}{1}}
1401 \DeclareCaptionOptionNoValue*{ignoreLTcapwidth}{%
1402   \caption@WarningNoLine{Obsolete option 'ignoreLTcapwidth' ignored}}

```

2.11 Obsolete caption v3.0 options

```

1403 \DeclareCaptionOption*{caption}{%
1404   \caption@setbool{temp}{#1}%
1405   \caption@ifbool{temp}{}{%
1406     \caption@Error{%
1407       The package option 'caption=#1' is obsolete.\MessageBreak
1408       Please pass this option to the subfig package instead\MessageBreak
1409       and do *not* load the caption package anymore}}}

```

2.12 fltpage package support options

With these options is controlled where the list-of entry and `\ref` resp. `\pageref` or `\autoref` will link to. Defaults are `FPlist=caption` and `FPref=figure` which is inconsistent, but compatible to the usual behaviour of the `fltpage` package.

```

1410 \DeclareCaptionOption{FPlist}[1]{\caption@setFPoption{list}{#1}}
1411 \DeclareCaptionOption{FPref}[1]{\caption@setFPoption{ref}{#1}}
1412 \@onlypreamble@key{caption}{FPlist}
1413 \@onlypreamble@key{caption}{FPref}
1414 \newcommand*\caption@setFPoption[2]{%
1415   \edef\@tempa{\@car#2@nil}%
1416   \caption@setbool{FP#1cap}{\if c\@tempa 1\else 0\fi}}
1417 \@onlypreamble\caption@setFPoption
1418 \captionsetup{FPlist=caption,FPref=figure}

```

2.13 hyperref package support options

With `hycap=off` one can turn the `hycap` support off (default is on).

```

1419 \DeclareCaptionOption{hycap}[1]{\caption@setbool{hycap}{#1}}
1420 \DeclareCaptionOption{hycapSPACE}{\def\caption@hycapSPACE{#1}}
1421 \captionsetup{hycap=1,hycapSPACE=.5\baselineskip}

```

2.14 Processing of options

```
1422 \caption@ProcessOptions*{caption}
```

2.15 Option ‘type’, \captionof and \ContinuedFloat

```
1423 \caption@AtBeginDocument{%
1424   \DeclareCaptionOption{type}{\caption@settype{#1}}
1425   \DeclareCaptionOption{type*}{\caption@settype*{#1}}}
```

Important Note: Like \captionof the option type= should only be used inside a group, box, or environment and does not check if the argument is a valid floating environment or not.

```
\caption@settype \caption@settype*{<type>}
```

sets \@capttype and executes the options associated with it (using \caption@setoptions). Furthermore we check \currentgrouplevel (if avail), redefine \@currentlabel so a \label before \caption will result in a hint instead of a wrong reference, and use the macro \caption@settypehook (which will be used by our float package support).

The non-starred version sets a hyperref anchor additionally (if hypcap=true and the hypcap package is not loaded).

```
1426 \newcommand*\caption@settype{%
1427   \caption@@settype{}}

1428 \newcommand*\caption@@settype[1]{%
1429   \caption@teststar{\caption@@@settype{#1}}{\@firstoftwo\@secondoftwo}

1430 \newcommand*\caption@@@settype[3]{%
1431   % #1 = "" or "sub"
1432   % #2 = \@firstoftwo in star form, \@secondoftwo otherwise
1433   % #3 = <type>, e.g. "figure" or "table"
1434   \caption@Debug{type=#3}%
1435   \caption@checkgrouplevel{#1}%
1436   \captionsetup{#1type#2*\@empty=...}#2{ or \@backslashchar#1captionof}{}}%

1437   \@ifundefined{@#1capttype}{}{%
1438     \edef\@tempa{#3}%
1439     \expandafter\ifx\csname @#1capttype\endcsname\@tempa \else
1440       \caption@iftypewarning{\caption@Warning{%
1441         Caption type was already set to `#3'\MessageBreak}}{}%
1442       \fi}%
1443   \expandafter\edef\csname @#1capttype\endcsname{#3}%
1444   \let\caption@iftypewarning\caption@ifdebug

1445   \@nameuse{#1caption@settypehook}%
1446   \expandafter\caption@setoptions\csname @#1capttype\endcsname

1447   \let\caption@ifrefstepcounter\@secondoftwo
1448   #2{}{%
1449     \let\@currentlabel\caption@undefinedlabel
1450     % \let\@currentHlabel\@undefined
1451     \ifx\caption@ORI@label\@undefined
1452       \let\caption@ORI@label\label
```

```

1453     \def\label{\caption@withoptargs\caption@xlabel}%
1454     \fi
1455     \caption@start}}

```

`\caption@settypehook` Hook, will be extended by our float package support.

```

1456 \newcommand*\caption@settypehook{}

```

`\caption@iftype` Since we often need to check if `\@capttype` is defined (means: we are inside a floating environment) this helper macro was introduced.

```

1457 \newcommand*\caption@iftype{%
1458   \@ifundefined{\@capttype}{\let\@capttype\@undefined\@secondoftwo}\@firstoftwo}

```

`\caption@checkgrouplevel` Checks if `\captionsetup{type=...}` or `\caption` is done inside a group or not – in the latter case an error will be issued. (needs $\mathcal{E}$ -TeX)

```

1459 \begingroup\expandafter\expandafter\expandafter\endgroup
1460 \expandafter\ifx\csname currentgrouplevel\endcsname\relax
1461   \caption@Debug{TeX engine: TeX}
1462   \let\caption@checkgrouplevel\@gobbletwo
1463 \else
1464   \caption@Debug{TeX engine: e-TeX}
1465   \newcommand*\caption@checkgrouplevel[2]{%
1466     \@ifundefined{#1caption@grouplevel}{%
1467       \@ifundefined{caption@grouplevel}{\let\caption@grouplevel\z@}{}}%
1468     \ifnum\currentgrouplevel>\caption@grouplevel\relax
1469       \expandafter\edef\csname #1caption@grouplevel\endcsname{%
1470         \the\currentgrouplevel}%
1471     \else
1472       \caption@Error{\string#2\MessageBreak outside box or environment}%
1473     \fi
1474   }{}}
1475 \fi

```

`\caption@undefinedlabel` This label will be used for `\currentlabel` inside (floating) environments as default. (see above)

```

1476 \newcommand*\caption@undefinedlabel{??}

```

`\caption@xlabel` The new code of `\label` inside floating environments. `\label` will be redefined using `\caption@withoptargs`, so #1 are the optional arguments (if any), and #2 is the mandatory argument here.

```

1477 \newcommand*\caption@xlabel[2]{%
1478   \ifx\@currentlabel\caption@undefinedlabel
1479     \caption@Warning{\noexpand\label before \noexpand\caption ignored}%
1480   \else
1481     \caption@ORI@label#1{#2}%
1482   \fi}

```

```

\captionof{\captionof{\langle type \rangle}[\langle lst.entry \rangle]{\langle heading \rangle}
\captionof*[\langle lst.entry \rangle]{\langle heading \rangle}

```

Note: This will be defined with `\AtBeginDocument` so `\usepackage{caption,capt-of}` will still work. (Compatibility to *v1.x*)

```
1483 \caption@AtBeginDocument{%
1484   \def\captionof{\caption@teststar\caption@of{\caption*}\caption}}
1485 \newcommand*\caption@of[2]{\caption@settype*{#2}#1}
```

```
\captionlistentry \captionlistentry[<float type>]{<list entry>}
1486 \newcommand*\captionlistentry{%
1487   \@testopt\caption@listentry\@capttype}
1488 \def\caption@listentry[#1]#2{%
1489   \caption@refstepcounter{#1}%
1490   \caption@makecurrent{#1}{#2}%
1491   \caption@addcontentsline{#1}{#2}}
```

```
\ContinuedFloat \ContinuedFloat
```

This mainly decrements the appropriate counter and increments the continuation counter instead. Furthermore we set `\caption@resetContinuedFloat` to `\@gobble` so the continuation counter will not be reset to zero inside `\caption@refstepcounter`. Please forget about the optional argument, it was never working well, is incompatible to the subfig package, but is still there for compatibility reasons.

Note: The definition of `\ContinuedFloat` itself is compatible to the one inside the subfig package, except for the optional argument.

```
1492 \def\ContinuedFloat{%
1493   \@ifnextchar[\@ContinuedFloat{%
1494     \caption@iftype
1495       {\addtocounter\@capttype\m@ne
1496        \caption@ContinuedFloat\@capttype}%
1497     {\caption@Error{\noexpand\ContinuedFloat outside float}}}}
1498 \def\caption@ContinuedFloat#1{%
1499   \edef\@tempa{#1}%
1500   \ifx\@tempa\caption@CFtype \else
1501     \caption@Error{Continued `#1' after '\caption@CFtype'}%
1502   \fi
1503   \stepcounter{ContinuedFloat}%
1504   \caption@@ContinuedFloat{#1}%
1505   \caption@setoptions{ContinuedFloat}%
1506   \caption@setoptions{continued#1}%
1507   \let\caption@resetContinuedFloat\@gobble}
1508 \def\@ContinuedFloat[#1]{\addtocounter{#1}\m@ne}
1509 \newcounter{ContinuedFloat}
1510 \newcommand*\caption@CFtype{??}
```

```
\caption@@ContinuedFloat \caption@@ContinuedFloat{<type>}
```

Usually this do nothing but this changes if the `hyperref` package is loaded. (See `hyperref` package support for details.)

```
1511 \newcommand*\caption@@ContinuedFloat[1]{}%
```

`\caption@resetContinuedFloat` `\caption@resetContinuedFloat{<type>}`
 If a continuation counter is defined, we reset it. (This one will be called inside
`\@caption`.)

```

1512 \newcommand*\caption@resetContinuedFloat[1]{%
1513   \@stpet{ContinuedFloat}\xdef\caption@CFtype{#1}}

```

2.16 Internal helpers

`\caption@refstepcounter` Resets the continuation counter, increments the float (i.e. figure or table) counter, and sets the `refstepcounter` flag.

```

1514 \newcommand*\caption@refstepcounter[1]{%
1515   \caption@resetContinuedFloat{#1}%
1516   \caption@@refstepcounter{#1}%
1517   \let\caption@ifrefstepcounter\@firstoftwo}

1518 \newcommand*\caption@@refstepcounter{\refstepcounter}
1519 \let\caption@ifrefstepcounter\@secondoftwo

```

`\caption@dblarg` A `\relax` was added compared to `\@dblarg` so `\caption{ }` will be expanded to `\caption[\relax]{ }` (and not to `\caption[]{ }`).

```

1520 \newcommand\caption@dblarg[1]{\kernel@ifnextchar[{\#1}{\caption@xdblarg{#1}}}%
1521 \newcommand\caption@xdblarg[2]{#1[{\#2\relax}]{\#2}}%

```

`\caption@begin` Our handling of `\caption` will always be surrounded by `\caption@begin` (or `\caption@beginex`) and `\caption@end`.
`\caption@begin{<type>}` performs these tasks:

1. Start a new group.
2. Define `\fnum{<type>}` if the caption label format is set to non-default.
3. Override the `position=` setting, if necessary. (for example if set to `auto` or used inside a `supertabular`)

```

1522 \newcommand*\caption@begin[1]{%
1523   \begin{group}
1524     \caption@setfnum{#1}%
1525     \caption@fixposition
1526     \global\let\caption@fixedposition\caption@position}

```

`\caption@beginex` `\caption@beginex{<type>}{<list entry>}{<heading>}`
 performs the same tasks as `\caption@begin` and additionally:

4. Make an entry in the list-of-whatever.
5. Set `\caption@ifempty` according argument `<heading>`.

```

1527 \newcommand\caption@beginex[3]{%
1528   \caption@begin{#1}%
1529   \caption@addcontentsline{#1}{#2}%
1530   \caption@ifempty{#3}{} }

```

`\caption@end` `\caption@end` closes the group.

```
1531 \newcommand*\caption@end{%
1532   \endgroup
1533   \let\caption@position\caption@fixedposition}
```

`\caption@setfnum` `\caption@setfnum{<type>}`
 redefines `\fnum@<type>` according the caption label format set with `labelformat=`.
 But if `labelformat=default` is set, `\fnum@<type>` will not be overwritten by us.

```
1534 \newcommand*\caption@setfnum[1]{%
1535   \@ifundefined{fnum@#1}{\iftrue}{\ifx\caption@lfmt\caption@lfmt@default\else}%
1536   \@namedef{fnum@#1}{\caption@fnum{#1}}%
1537   \fi}
```

`\caption@addcontentsline` If `\sf@ifpositiontop` is defined, we extend `\caption@addcontentsline` so it handles sub-captions, too.

```
1538 \caption@AtBeginDocument*{%
1539   \@ifundefined{sf@ifpositiontop}{}{%
1540     \let\caption@kernel@addcontentsline\caption@addcontentsline
1541     \renewcommand*\caption@addcontentsline[2]{%
1542       \sf@ifpositiontop{\@listsubcaptions{#1}}{}%
1543       \caption@kernel@addcontentsline{#1}{#2}%
1544       \sf@ifpositiontop{}{\@listsubcaptions{#1}}}%
1545   }}
```

`\caption@boxrestore` The original code (from `latex/base/ltboxes.dtx`):

```
\def\@parboxrestore{\@arrayparboxrestore\let\\\@normalcr}
\def\@arrayparboxrestore{%
  \let\if@nbreak\iffalse
  \let\if@noskipsec\iffalse
  \let\par\@par
  \let\-\@dischyph
  \let'\@acci\let'\@accii\let\=\@acciii
  \parindent\z@ \parskip\z@skip
  \everypar{}%
  \linewidth\hsize
  \@totalleftmargin\z@
  \leftskip\z@skip \rightskip\z@skip \@rightskip\z@skip
  \parfillskip\@flushglue \lineskip\normallineskip
  \baselineskip\normalbaselineskip
  \sloppy}
```

This one will be used by `\@caption` instead of `\@parboxrestore`.

```
1546 \newcommand*\caption@boxrestore{%
1547   \let\if@nbreak\iffalse
1548   \let\if@noskipsec\iffalse
1549   \let\par\@par
1550   \let\-\@dischyph
1551   \let'\@acci\let'\@accii\let\=\@acciii
```

```

1552 \parindent\z@ \parskip\z@skip
1553 \everypar{}%
1554 % \linewidth\hsize
1555 % \@totalleftmargin\z@
1556 \leftskip\z@skip \rightskip\z@skip \@rightskip\z@skip
1557 \parfillskip\@flushglue \lineskip\normallineskip
1558 \baselineskip\normalbaselineskip
1559 \sloppy
1560 \let\\\@normalcr
1561 }

```

`\caption@normalsize` This one will be used by `\@caption` instead of `\normalsize`.
Its code is equivalent to

```
\caption@font{normal}%
```

but executes faster (since the starred form of `\caption@font` does not use `\setkeys` internally).

```

1562 \newcommand*\caption@normalsize{%
1563 \caption@font*\KV@caption@fnt@normal\@unused}}

```

`\caption@setfloatcapt` Needed for support of the float package, where the caption will not be typeset directly, but caught in a `\vbox` called `\@floatcapt` instead.

```
1564 \let\caption@setfloatcapt\@firstofone
```

`\caption@makecurrent` All these are needed for support of the hyperref package.

```

\caption@@anchor 1565 \newcommand*\caption@makecurrent[2]{}
\caption@start 1566 \let\caption@@anchor\@firstofone
\caption@@start 1567 \let\caption@start\relax
\caption@freezeHref 1568 \let\caption@@start\relax
\caption@defrostHref 1569 \let\caption@freezeHref\relax
1570 \let\caption@defrostHref\relax

```

2.17 `\caption`, `\@caption`, and `\@makecaption`

We only redefine `\caption` and `\@caption` if the current definitions are well known, so documents written in the old (caption package *v1.x*) days (where `\caption` & `\@caption` were not redefined by us) will still compile fine. For example the usage of the `captcont` package, which brings it's own definition of `\caption*`, was quite common these days.

Some packages (like the `hyperref` package for example) redefines `\caption` and `\@caption`, too. So we have to use `\AtBeginDocument` here, so we can make sure our definition is the one which will be valid at last.

```

1571 \caption@AtBeginDocument{%
1572 \caption@setbool{incompatible}{0}%
1573 \caption@CheckCommand\caption{%
1574 % ltfloating.dtx [2002/10/01 v1.1v LaTeX Kernel (Floats)]

```

```

1575 \def\caption{%
1576     \ifx\@capttype\@undefined
1577         \@latex@error{\noexpand\caption outside float}\@ehd
1578         \expandafter\@gobble
1579     \else
1580         \refstepcounter\@capttype
1581         \expandafter\@firstofone
1582     \fi
1583     {\@dblarg{\@caption\@capttype}}%
1584 }}%

1585 \caption@CheckCommand\caption{%
1586     % beamerbaselocalstructure.sty,v 1.53 2007/01/28 20:48:21 tantau
1587     \def\caption{
1588         \ifx\@capttype\@undefined
1589             \@latex@error{\noexpand\caption outside figure or table}\@ehd
1590             \expandafter\@gobble
1591         \else
1592             \refstepcounter\@capttype
1593             \expandafter\@firstofone
1594         \fi
1595         {\@dblarg{\@caption\@capttype}}%
1596     }%

1597 \caption@CheckCommand\caption{%
1598     % float.sty [2001/11/08 v1.3d Float enhancements (AL)]
1599     \renewcommand\caption{%
1600         \ifx\@capttype\@undefined
1601             \@latex@error{\noexpand\caption outside float}\@ehd
1602             \expandafter\@gobble
1603         \else
1604             \refstepcounter\@capttype
1605             \let\@tempf\@caption
1606             \expandafter\ifx\csname @float@c@\@capttype\endcsname\relax\else
1607                 \expandafter\expandafter\let
1608                 \expandafter\@tempf\csname @float@c@\@capttype\endcsname
1609             \fi
1610         \fi
1611         \@dblarg{\@tempf\@capttype}}}%

1612 \caption@CheckCommand\caption{%
1613     % hyperref.sty [2007/02/27 v6.75t Hypertext links for LaTeX]
1614     % hyperref.sty [2007/04/09 v6.76a Hypertext links for LaTeX]
1615     % hyperref.sty [2007/06/12 v6.76h Hypertext links for LaTeX]
1616     \def\caption{%
1617         \ifx\@capttype\@undefined
1618             \@latex@error{\noexpand\caption outside float}\@ehd
1619             \expandafter\@gobble
1620         \else
1621             \H@refstepcounter\@capttype
1622             \@ifundefined{fst@\@capttype}{%
1623                 \let\Hy@tempa\@caption

```

```

1624         }{%
1625         \let\Hy@tempa\Hy@float@caption
1626         }%
1627         \expandafter\@firstofone
1628         \fi
1629         {\@dblarg{\Hy@tempa\@capttype}}%
1630     } }%

1631 \caption@CheckCommand\caption{%
1632 % hyperref.sty [2007/08/05 v6.76j Hypertext links for LaTeX]
1633 \def\caption{%
1634     \ifx\@capttype\@undefined
1635     \latex@error{\noexpand\caption outside float}\@ehd
1636     \expandafter\@gobble
1637     \else
1638     \H@refstepcounter\@capttype
1639     \let\Hy@tempa\@caption
1640     \@ifundefined{float@caption}{%
1641     }{%
1642     \expandafter\ifx\csname @float@c@%\@capttype\endcsname\float@caption
1643     \let\Hy@tempa\Hy@float@caption
1644     \fi
1645     }%
1646     \expandafter\@firstofone
1647     \fi
1648     {\@dblarg{\Hy@tempa\@capttype}}%
1649     } }%

1650 \caption@IfCheckCommand{}{%
1651 \caption@Info{%
1652     Incompatible package detected (regarding \string\caption).\MessageBreak
1653     \string\caption\space=\space\meaning\caption}%
1654 \caption@setbool{incompatible}{1}}%

1655 \caption@CheckCommand\@caption{%
1656 % ltfloating.dtx [2002/10/01 v1.1v LaTeX Kernel (Floats)]
1657 \long\def\@caption#1[#2]#3{%
1658     \par
1659     \addcontentsline{\csname ext@#1\endcsname}{#1}%
1660     {\protect\numberline{\csname the#1\endcsname}{\ignorespaces #2}}%
1661     \begingroup
1662     \@parboxrestore
1663     \if@minipage
1664     \setminipage
1665     \fi
1666     \normalsize
1667     \makecaption{\csname fnum@#1\endcsname}{\ignorespaces #3}\par
1668     \endgroup}%

1669 \caption@CheckCommand\@caption{%
1670 % beamerbaselocalstructure.sty,v 1.53 2007/01/28 20:48:21 tantau
1671 \long\def\@caption#1[#2]#3{% second argument ignored

```

```

1672     \par\nobreak
1673     \begingroup
1674     \@parboxrestore
1675     \if@minipage
1676     \setminipage
1677     \fi
1678     \beamer@makecaption{#1}{\ignorespaces #3}\par\nobreak
1679     \endgroup}}%

1680 % \caption@CheckCommand\float@caption{%
1681 %   float.sty [2001/11/08 v1.3d Float enhancements (AL)]
1682 %   \long\def\float@caption#1[#2]#3{%
1683 %     \addcontentsline{\@nameuse{ext@#1}}{#1}%
1684 %     {\protect\numberline{\@nameuse{the#1}}{\ignorespaces #2}}
1685 %     \global\setbox\@floatcapt\vbox\bgroup\@parboxrestore
1686 %     \normalsize\@fs@capt{\@nameuse{fnum@#1}}{\ignorespaces #3}%
1687 %     \@ifnextchar[{\float@ccon}{\egroup}}%
1688 %   \long\def\float@ccon[#1]#1\par\egroup}}%

1689 \caption@CheckCommand\@caption{%
1690 % hyperref.sty [2007/02/27 v6.75t Hypertext links for LaTeX]
1691 \long\def\@caption#1[#2]#3{%
1692   \hyper@makecurrent{\@cuptype}%
1693   \def\@currentlabelname{#2}%
1694   \par\addcontentsline{\csname ext@#1\endcsname}{#1}{%
1695     \protect\numberline{\csname the#1\endcsname}{\ignorespaces #2}%
1696   }%
1697   \begingroup
1698   \@parboxrestore
1699   \if@minipage
1700   \setminipage
1701   \fi
1702   \normalsize
1703   \@makecaption{\csname fnum@#1\endcsname}{%
1704     \ignorespaces
1705     \ifHy@nesting
1706       \hyper@@anchor{\@currentHref}{#3}%
1707     \else
1708       \Hy@raisedlink{\hyper@@anchor{\@currentHref}{\relax}}#3%
1709     \fi
1710   }%
1711   \par
1712   \endgroup
1713 }}%

1714 \caption@CheckCommand\@caption{%
1715 % hyperref.sty [2007/04/09 v6.76a Hypertext links for LaTeX]
1716 % hyperref.sty [2007/06/12 v6.76h Hypertext links for LaTeX]
1717 % hyperref.sty [2007/08/05 v6.76j Hypertext links for LaTeX]
1718 \long\def\@caption#1[#2]#3{%
1719   \expandafter\ifx\csname if@capstart\expandafter\endcsname
1720     \csname iftrue\endcsname

```

```

1721     \global\let\@currentHref\hc@currentHref
1722   \else
1723     \hyper@makecurrent{\@captype}%
1724   \fi
1725   \def\@currentlabelname{#2}%
1726   \par\addcontentsline{\csname ext@#1\endcsname}{#1}{%
1727     \protect\numberline{\csname the#1\endcsname}{\ignorespaces #2}%
1728   }%
1729   \beginngroup
1730     \@parboxrestore
1731     \if@minipage
1732       \setminipage
1733     \fi
1734     \normalsize
1735     \expandafter\ifx\csname if@capstart\expandafter\endcsname
1736       \csname iftrue\endcsname
1737       \global\@capstartfalse
1738       \makecaption{\csname fnum@#1\endcsname}{\ignorespaces#3}%
1739     \else
1740       \makecaption{\csname fnum@#1\endcsname}{%
1741         \ignorespaces
1742         \ifHy@nesting
1743           \hyper@@anchor{\@currentHref}{#3}%
1744         \else
1745           \Hy@raisedlink{\hyper@@anchor{\@currentHref}{\relax}}#3%
1746         \fi
1747       }%
1748     \fi
1749     \par
1750   \endgroup
1751 } }%

1752 \caption@CheckCommand\@caption{%
1753   % nameref.sty [2006/12/27 v2.28 Cross-referencing by name of section]
1754   \long\def\@caption#1[#2]{%
1755     \def\@currentlabelname{#2}%
1756     \NR@@caption{#1}[{#2}]%
1757   } }%

1758 \caption@CheckCommand\@caption{%
1759   % subfigure.sty [2002/07/30 v2.1.4 subfigure package]
1760   \long\def\@caption#1[#2]#3{%
1761     \ifundefined{if#1topcap}%
1762       {\subfig@oldcaption{#1}[{#2}]{#3}}%
1763     {\@nameuse{if#1topcap}%
1764       \@listsubcaptions{#1}%
1765       \subfig@oldcaption{#1}[{#2}]{#3}%
1766     \else
1767       \subfig@oldcaption{#1}[{#2}]{#3}%
1768       \@listsubcaptions{#1}%
1769     \fi}}%

```

```

1770 \caption@CheckCommand\@caption{%
1771   % subfig.sty [2005/06/28 ver: 1.3 subfig package]
1772   \def\@caption{\caption@}%
1773 % \long\def\caption@#1[#2]#3{%
1774 %   \@ifundefined{caption@setfloattype}%
1775 %     \caption@settype
1776 %     \caption@setfloattype
1777 %     \@capttype
1778 %   \sf@ifpositiontop{%
1779 %     \@listsubcaptions{#1}%
1780 %     \sf@old@caption{#1}[{#2}]{#3}%
1781 %   }{%
1782 %     \sf@old@caption{#1}[{#2}]{#3}%
1783 %     \@listsubcaptions{#1}%
1784 %   }}%
1785   }%

1786 \caption@IfCheckCommand{}{%
1787   \caption@Info{%
1788     Incompatible package detected (regarding \string\@caption).\MessageBreak
1789     \string\@caption\space=\space\meaning\@caption}%
1790   \caption@setbool{incompatible}{1}}%

```

The option compatibility= will override the compatibility mode.

```

1791 \@ifundefined{caption@ifcompatibility}%
1792   {\let\caption@ifcompatibility\caption@ifincompatible
1793   \let\@tempa\@firstofone}%
1794   {\let\@tempa\@gobble}% suppress warning
1795 \caption@ifcompatibility{%
1796   \@tempa{%
1797     \caption@WarningNoLine{%
1798       \noexpand\caption will not be redefined since it's already\MessageBreak
1799       redefined by a document class or package which is\MessageBreak
1800       unknown to the caption package.\MessageBreak
1801       As a result some features will not work as expected}%
1802     }%

```

\ContinuedFloat is not supported in compatibility mode.

```

1803 \renewcommand*\caption@ContinuedFloat[1]{%
1804   \caption@Error{Not available in compatibility mode}}%

```

\caption@start is not supported in compatibility mode.

```

1805 \caption@AtBeginDocument*{%
1806   \let\caption@start\relax
1807   \@ifundefined{caption@ORI@capstart}{}{%
1808     \caption@Debug{%
1809       Restore hypcap definition of \string\capstart\@gobble}%
1810     \let\capstart\caption@ORI@capstart}%
1811   \@ifundefined{caption@ORI@float@makebox}{}{%
1812     \caption@Debug{%

```

```

1813         Restore hyperref redefinition of \string\float@makebox\@gobble}%
1814         \let\float@makebox\caption@ORI@float@makebox}%
1815     }%

\caption@star  We define \caption@star here, too, so it's defined but does not make any harm.

1816     \newcommand*\caption@star[2]{#1#2}%

1817 }{%

1818     \caption@ifincompatible{%
1819         \caption@WarningNoLine{%
1820             Forced redefinition of \noexpand\caption since the\MessageBreak
1821             unsupported(!) package option 'compatibility=false'\MessageBreak
1822             was given. Please fasten your seat belts, as we may\MessageBreak
1823             experience turbulence}%
1824     }{}%

\caption  Here comes our definition of \caption and \caption*. Beside the support of the
starred variant this code was adapted to the various packages we support. We are using
\caption@dblarg instead of \@dblarg so \caption{} (with an empty arg.) will
produce a list-of entry, but \caption[]{} won't.

1825     \def\caption{%
1826         \caption@iftype
1827         {\caption@checkgrouplevel\@empty\caption
1828          \caption@star
1829           {\caption@refstepcounter\@capttype}%
1830           {\caption@dblarg{\@caption\@capttype}}}%
1831         {\caption@Error{\noexpand\caption outside float}}}%

\caption@star  A helper macro which processes the optional * after \caption.
Note: We set \caption@startrue globally so it works with the sidecap package, too.

1832     \newcommand*\caption@star[2]{%
1833         \@ifstar{\global\caption@startrue#2[]}{#1#2}}%

\@caption  As above, our version has been adapted to the packages we support. Addition-
ally our code is nested by \caption@beginex & \caption@end instead of
\begin group & \end group. Furthermore we use \caption@boxrestore in-
stead of \@parboxrestore so this code also works correctly inside list-based environ-
ments like wide & addmargin. (This, and the fact that we use \linewidth instead
of \hsize inside \@makecaption, solves LATEX PR latex/2472.)

1834     \long\def\@caption#1[#2]#3{%
1835         \ifcaption@star \else
1836             \caption@makecurrent{#1}{#2}%
1837             \caption@ifhyperc{\caption@@start}{}%
1838         \fi
1839         \par
1840         \caption@beginex{#1}{#2}{#3}%
1841         \caption@setfloatcapt{%

```

```

1842         \caption@boxrestore
1843         \caption@normalsize
1844         \ifcaption@star
1845             \let\caption@@anchor\@firstofone
1846         \fi
1847         \@makecaption{\csname fnum@#1\endcsname}%
1848                     {\ignorespaces\caption@@anchor{#3}}\par}%
1849     \caption@end}%

1850 }%

1851 \caption@AtBeginDocument*{%
1852     \let\caption@ORI@capstart\@undefined
1853     \let\caption@ORI@float@makebox\@undefined}%

\@xfloat    We redefine \@xfloat so inside floating environments our type-specific options will be
            used, a hyperref anchor will be set etc.

1854     \let\caption@ORI@xfloat\@xfloat
1855     \def\@xfloat#1[#2]{%
1856         \caption@ORI@xfloat{#1}[#2]%
1857         \caption@settype{#1}}%

1858 }

\@makecaption \@makecaption{\langle label \rangle}{\langle text \rangle}
            We do basically the same as the original code (from the standard LATEX document classes),
            but take care of the position= setting and use \caption@@make from the caption
            kernel to finally typeset the caption.

1859 \long\def\@makecaption#1#2{%
1860     \caption@iftop{\vskip\belowcaptionskip}{\vskip\abovecaptionskip}%
1861     \caption@@make{#1}{#2}%
1862     \caption@iftop{\vskip\abovecaptionskip}{\vskip\belowcaptionskip}}

```

2.18 Document class & Babel package support

2.18.1 The $\mathcal{A}\mathcal{M}\mathcal{S}$ & SMF classes

```

1863 \ifundefined{smf@makecaption}{}{\let\smf@makecaption\@makecaption}

```

2.18.2 The beamer class

```

1864 \ifclassloaded{beamer}{%
1865     \caption@Info{beamer document class}%

```

Since the beamer class do not offer a ‘list of figures’ we switch this support in the caption package off.

```

1866     \captionsetup{listof=false}
1867     \undefine@key{caption}{listof}
1868     \undefine@key{caption}{listofformat}
1869 }{}

```

2.18.3 The KOMA-Script classes

KOMA-Script contains the code `\AtBeginDocument{\let\scr@caption\caption}` so we need to update `\scr@caption` here, too.

```
1870 \@ifundefined{scr@caption}{}{%
1871   \caption@AtBeginDocument{\let\scr@caption\caption}}
```

2.18.4 The frenchb Babel option

Suppress “Package frenchb.1df Warning: The definition of `\@makecaption` has been changed, frenchb will NOT customize it.” (but only if we emulate this customization)

```
1872 \@ifundefined{caption@frenchb}{}{\caption@frenchb}
```

2.18.5 The frenchle/pro package

```
1873 \caption@AtBeginDocument{\@ifundefined{frenchTeXmods}{}{%
1874   \caption@Info{frenchle/pro package is loaded}%
1875   \let\captionfont@ORI\captionfont
1876   \let\captionlabelfont@ORI\captionlabelfont
1877   \let\@makecaption@ORI\@makecaption
```

If `\GOfrench` is defined as `\relax` all the re-definitions regarding captions have already been done, so we can do our patches immediately. Otherwise we must add our stuff to `\GOfrench`.

```
1878   \@ifundefined{GOfrench}%
1879     {\let\@tempa\@firstofone}%
1880     {\def\@tempa{\g@addto@macro\GOfrench}}%
1881   \@tempa{%
1882     \let\captionfont\captionfont@ORI
1883     \let\captionfont@ORI\@undefined
1884     \let\captionlabelfont\captionlabelfont@ORI
1885     \let\captionlabelfont@ORI\@undefined
1886     \let\@makecaption\@makecaption@ORI
1887     \let\@makecaption@ORI\@undefined
```

`\@cnORI` We update the definition of `\@cnORI` so it actually reflects our definition of `\caption`.

```
1888   \let\@cnORI\caption
```

`\@tablescaption` The frenchle/pro package sets `\caption` to `\@tablescaption` at `\begin{table}` for special treatment of footnotes. Therefore we have to patch `\@tablescaption` so `\caption*` will work inside the table environment.

```
1889   \let\caption@tcORI\@tablescaption
1890   \def\@tablescaption{\caption@star\relax\caption@tcORI}%
```

`\f@ffrench` `\f@tfrench` and `\f@tfrench` reflect `\fnum@figure` and `\fnum@table` when used in French mode. These contain additional code which typesets the caption separator `\captionseparator` instead of the usual colon. Because this breaks with our `\@makecaption` code we have to remove this additional code here.

```

1891 \let\eatDP\@undefined
1892 \let\@tempa\@empty
1893 \ifx\ffrench\fnun@figure
1894 \l@addto@macro\@tempa{\let\fnun@figure\ffrench}%
1895 \fi
1896 \ifx\ftfrench\fnun@table
1897 \l@addto@macro\@tempa{\let\fnun@table\ftfrench}%
1898 \fi
1899 \def\ffrench{\ifx\listoffigures\relax\else\figurename~\thefigure\fi}%
1900 \def\ftfrench{\ifx\listoftables\relax\else\tablename~\thetable\fi}%
1901 \@tempa

1902 }%
1903 }}

```

2.19 Package support

```

\caption@ifpackageloaded \caption@ifpackageloaded{<package>}[<version>]{<true>}{<false>}
Some kind of combination of \ifpackageloaded and \ifpackagelater. If
the <package> is not loaded yet, the check will be (re-)done \AtBeginDocument, so
the <package> could be loaded later on, too.

1904 \newcommand\caption@ifpackageloaded[1]{%
1905 \@testopt{\caption@ifpackageloaded{#1}}{}}
1906 \@onlypreamble\caption@ifpackageloaded

1907 \long\def\caption@ifpackageloaded#1[#2]#3#4{%
1908 \ifpackageloaded{#1}\@firstofone{%
1909 \caption@Debug{#1 package is not loaded (yet)\@gobble}%
1910 \caption@AtBeginDocument}{%
1911 \caption@ifpackageloaded{#1}[#2]{#3}{#4}}
1912 \@onlypreamble\caption@ifpackageloaded

1913 \newcommand\caption@ifpackageloaded[1]{%
1914 \@testopt{\caption@ifpackageloaded{#1}}{}}
1915 \@onlypreamble\caption@ifpackageloaded

1916 \long\def\caption@ifpackageloaded#1[#2]{%
1917 \ifpackageloaded{#1}{%
1918 \caption@Info{#1 package is loaded}%
1919 \ifpackagelater{#1}{#2}\@firstoftwo{%
1920 \caption@Error{%
1921 For a successful cooperation we need at least version\MessageBreak
1922 `#2' of package #1,\MessageBreak
1923 but only version\MessageBreak
1924 '\csname ver@#1.\@pkgextension\endcsname'\MessageBreak
1925 is available}%
1926 \@secondoftwo}%
1927 }{\@secondoftwo}}
1928 \@onlypreamble\caption@ifpackageloaded

```

`\caption@clearmargin` This macro will be used by some package support stuff where the usual margin setting is not welcome, e.g. in the `sidecap` package.

```

1929 \newcommand*\caption@clearmargin{%
1930   \setcaptionmargin\z@
1931   \let\caption@minmargin\@undefined}

1932 \caption@setbool{needfreeze}{0}
1933 \caption@AtBeginDocument*{%
1934   \caption@ifneedfreeze{%

```

`\caption@freeze` `\caption@freeze*`
Used by the `fltpage` & `sidecap` package support.

```

1935 \newcommand*\caption@freeze{%
1936   \caption@teststar\caption@@freeze\@gobble\@firstofone}%

1937 \newcommand*\caption@@freeze[1]{%
1938   \global\let\caption@SCcontinued\relax
1939   \global\let\caption@SCsetup\@undefined
1940   \global\let\caption@SClentry\@undefined
1941   \global\let\caption@SCtext\@undefined
1942   \global\let\caption@SClabel\@undefined

1943   \let\caption@ORI@ContinuedFloat\ContinuedFloat
1944   \def\ContinuedFloat{%
1945     \let\caption@ORI@setcounter\setcounter
1946     \let\caption@ORI@addtocounter\addtocounter
1947     \def\setcounter####1####2{\csname c####1\endcsname####2\relax}%
1948     \def\addtocounter####1####2{\advance\csname c####1\endcsname ####2\relax}%
1949     \caption@ORI@ContinuedFloat
1950     \global\let\caption@SCcontinued\caption@ORI@ContinuedFloat
1951     \let\setcounter\caption@ORI@setcounter
1952     \let\addtocounter\caption@ORI@addtocounter}%
1953   \def\captionsetup##1{%
1954     \caption@g@addto@list\caption@SCsetup{##1}%
1955     \caption@setup{##1}}%
1956   \let\caption@ORI\caption
1957   \def\caption{%
1958     \def\caption{\caption@Error{%
1959       Only one \noexpand\caption can be placed in this environment}}%
1960     \let\captionsetup\caption@setup
1961     \let\caption@@refstepcounter\caption@l@stepcounter
1962     \caption@ORI}%
1963   \long\def\@caption##1[##2]##3{%
1964     \gdef\caption@SClentry{##2}\gdef\caption@SCtext{##3}%
1965     \ignorespaces}%
1966   #1{% is \@gobble in star form, and \@firstofone otherwise
1967     \def\label##1{\gdef\caption@SClabel{##1}\ignorespaces}}%

1968   }%

```

`\caption@defrost` `\caption@defrost`

```

1969 \newcommand*\caption@defrost{%
1970   \ifx\caption@ORI@ContinuedFloat\@undefined
1971     \caption@SCcontinued
1972     \ifx\caption@SCsetup\@undefined \else
1973       \expandafter\captionsetup\expandafter{\caption@SCsetup}%
1974     \fi
1975     \ifx\caption@SCtext\@undefined \else
1976       \expandafter\expandafter\expandafter\caption
1977       \expandafter\expandafter\expandafter{%
1978       \expandafter\expandafter\expandafter%
1979       \expandafter\caption@SClentry\expandafter}\expandafter}%
1980     \expandafter{\caption@SCtext}%
1981   \fi
1982   \ifx\caption@SClabel\@undefined \else
1983     \expandafter\label\expandafter{\caption@SClabel}%
1984   \fi
1985   \else
1986     \caption@Error{Internal Error:\MessageBreak
1987       \noexpand\caption@defrost in same group as \string\caption@freeze}%
1988   \fi}%
1989 }{}%
1990 \caption@undefbool{needfreeze}}

```

2.19.1 The float package

The float package usually do not use the L^AT_EX kernel command `\caption` to typeset the caption but `\float@caption` instead. (`\caption` will only be used if the float is re-styled with `\restylefloat*`.)

The main two things `\float@caption` is doing different are:

- The caption will be typeset inside a `\savebox` called `\@floatcapt` so it can be placed above or below the float contents afterwards.
- `\@makecaption` will not be used to finally typeset the caption. Instead `\@fs@capt` will be used which definition is part of the float style. (Note that `\@fs@capt` will not typeset any vertical space above or below the caption; instead this space will be typeset by the float style code itself.)

```

1991 \caption@ifPackageLoaded{float}[2001/11/08 v1.3d]{%
1992   \@ifpackageloaded{floatrow}{%
1993     \caption@ifpackageloaded{floatrow}[2007/08/24 v0.2a]{}{}%
1994   }{}%

```

`\@float@setevery` `\@float@setevery{float type}` is provided by the float package; it's called every time a floating environment defined with `\newfloat` or `\restylefloat` begins. We use this hook to do some adaptations and to setup the proper caption style (if defined) and additional settings declared with `\captionsetup[float style]`.

```

1995 \let\caption@ORI@float@setevery\@float@setevery
1996 \def\@float@setevery#1{%
1997   \float@ifcaption{#1}{%

```

First of all we set the caption position to its proper value by converting `\@fs@iftopcapt` (which is part of a float style and controls where the caption will be typeset, above or below the float contents) to our `position=` setting. Since the spacing above and below the caption will be done by the float style and *not* by us this sounds quite useless. But in fact it isn't, since some packages based on the caption package (like the subfig package) could have an interest for this information and therefore use the `\caption@iftop` macro we provide in our kernel. Furthermore we need this information for ourself in `\captionof` which uses `\@makecaption` to finally typeset the caption with skips.

```
1998      \caption@setposition{\@fs@iftopcapt t\else b\fi}%
```

Afterward we redefine `\caption@setfloatcapt` (which will be used inside `\@caption`) so the caption will be set inside the box `\@floatcapt`, without extra vertical space.

```
1999      \renewcommand\caption@setfloatcapt{%
2000      \let\@makecaption\caption@@make
2001      \global\setbox\@floatcapt\vbox}%
```

To allow different caption styles for different float styles we also determine the current float style (e.g. 'ruled') and select a caption style (and additional settings) with the same name, if defined.

```
2002      \float@getstyle\float@style{#1}%
2003      \caption@setstyle*\float@style
2004      \caption@setoptions\float@style
2005      }{}%
2006      \caption@freezeHref % will be defrosted in \float@makebox
2007      \caption@ORI@float@setevery{#1}}%
```

`\caption@settypehook` $\LaTeX$ and almost every other packages use `\langle type \rangle` name to provide a macro for the type resp. environment name – for example the command `\figurename` will usually contain the name of the floating environment figure:

```
\newcommand\figurename{Figure}
```

But the float package doesn't follow this common naming convention: For floats defined with `\newfloat` it uses `\fname@⟨type⟩` instead, which breaks with our code (and with `\autoref` and some other things as well). So we have to map the float package name to the common one here.

Note: If the float was not defined with `\newfloat` but with `\restylefloat` instead, `\fname@⟨type⟩` is not defined.

```
2008      \g@addto@macro\caption@settypehook{%
2009      \expandafter\ifx\csname #1name\endcsname\relax
2010      \expandafter\let\csname #1name\endcsname\expandafter\endcsname
2011      \csname fname@#1\endcsname
2012      \fi}%
```

`\fs@plaintop` `\fs@boxed` Since the float styles `plaintop` and `boxed` don't use `\abovecaptionskip` which could be set with `skip=` (`plaintop` uses `\belowcaptionskip` instead of `\abovecaptionskip`, and `boxed` uses a fixed space of 2pt) we patch the according float style macros here to change this.

```

2013 \g@addto@macro\fs@plaintop{\def\@fs@mid{\vspace\abovecaptionskip\relax}}%
2014 \g@addto@macro\fs@boxed{\def\@fs@mid{\kern\abovecaptionskip\relax}}%

```

```
\float@ifstyle \float@ifstyle{<type>}{<if-clause>}{<else-clause>}
```

Checks if the given *<type>* (e.g. figure) is associated with a float style (e.g. boxed).

```

2015 \providecommand*\float@ifstyle[1]{%
2016   \expandafter\ifx\csname fst@#1\endcsname\relax
2017     \expandafter\@secondoftwo
2018   \else
2019     \expandafter\@firstoftwo
2020   \fi}%

```

```
\float@getstyle \float@getstyle{<cmd>}{<type>}
```

Determining the float style is not so easy because the only hint provided by the float package is the macro `\fst@<float type>` which points to the macro which represents the float style. So for example after

```

\floatstyle{ruled}
\newfloat{Program}{tbp}{lop}

```

`\fst@Program` will be defined as

```
\def\fst@Program{\fs@ruled} .
```

So here is what we do: We make the first level expansion of `\fst@<float type>` a string so we can gobble the first four tokens (= `\fs@`), so only the the name of the float style is left.

TODO: We need to convert the catcodes here.

```

2021 \providecommand*\float@getstyle[2]{%
2022   \edef#1{%
2023     \noexpand\expandafter\noexpand\@gobblefour\noexpand\string
2024     \expandafter\expandafter\expandafter\noexpand
2025     \csname fst@#2\endcsname}%
2026   \edef#1{#1}%
2027   \caption@Debug{floatstyle{#2} = '#1'}}%

```

```
\float@setstyle \float@setstyle{<type>}{<style>}
```

Sets or changes the float style associated with *<type>*.

```

2028 \providecommand*\float@setstyle[2]{%
2029   \expandafter\edef\csname fst@#1\endcsname{%
2030     \expandafter\noexpand\csname fs@#2\endcsname}}%

```

```
\float@dostyle \float@dostyle{<type>}
```

```

2031 \providecommand*\float@dostyle[1]{%
2032   \@nameuse{fst@#1}\@float@setevery{#1}}%

```

```
\float@ifcaption \float@ifcaption{<type>}{<if-clause>}{<else-clause>}
```

Here we determine if the user has used `\newfloat` resp. `\restylefloat`, or `\restylefloat*`. This is quite easy: If `\@float@c@<captype>` is the same as

\float@caption, the user has used \newfloat or \restylefloat, otherwise we assume he has used \restylefloat*. (This test will fail if some package re-defines \float@caption, so we have to assume that there is no one.)

```

2033 \providecommand*\float@ifcaption[1]{%
2034   \expandafter\ifx\csname @float@c@#1\endcsname\float@caption
2035   \expandafter\@firstoftwo
2036   \else
2037   \expandafter\@secondoftwo
2038   \fi}%

2039 } }{%
2040 \providecommand*\float@ifstyle[1]{\@secondoftwo}%
2041 \providecommand*\float@ifcaption[1]{\@secondoftwo}%
2042 % \clearcaptionsetup{boxed}% used by the floatrow package?
2043 }

```

The skip between ‘boxed’ floats and their caption defaults to 2pt.

```

2044 \captionsetup{boxed}{skip=2pt} % do not issue a warning when not used

```

To emulate the ‘ruled’ definition of \@fs@capt we provide a caption style ‘ruled’ with appropriate options. But if the package option ruled was specified, we setup some caption parameters to emulate the behavior of the caption package v1.x option ruled instead, i.e., the current caption settings will be used, but without margin and without ‘single-line-check’.

```

2045 \caption@ifbool{ruled}{%
2046   \captionsetup{ruled}{margin=0pt,minmargin=0,singlelinecheck=0}%
2047 }{%
2048   \DeclareCaptionStyle{ruled}{labelfont=bf,labelsep=space,strut=0}%
2049 }
2050 \caption@undefbool{ruled}

```

2.19.2 The floatflt package

```

2051 \caption@ifPackageLoaded{floatflt}[1996/02/27 v1.3]{%

```

\floatingfigure We patch \floatingfigure so \caption@floatflt will be used.

```

2052 \let\caption@ORI@floatingfigure\floatingfigure
2053 \def\floatingfigure{%
2054   \caption@floatflt{figure}%
2055   \caption@ORI@floatingfigure}%

```

\floatingtable Same with \floatingtable...

```

2056 \let\caption@ORI@floatingtable\floatingtable
2057 \def\floatingtable{%
2058   \caption@floatflt{table}%
2059 % \caption@setautoposition b%
2060   \caption@ORI@floatingtable}%

```

\caption@floatflt Here we do two things:

1. We use `\caption@setoptions{floating<type>}` so `\captionsetup[floating<type>]{...}` is supported.
2. `\linewidth` must be set correctly. Usually this is done by `\@parboxrestore` inside `\@caption`, but since we use `\@caption@boxrestore` we have to map this to `\@parboxrestore` instead.

```

2061 \newcommand*\caption@floatflt[1]{%
2062   \caption@settype{#1}%
2063   \caption@clearmargin
2064   \caption@setoptions{floating#1}%
2065   \let\caption@boxrestore\@parboxrestore}%
2066 }{}

```

2.19.3 The fltpage package

```

2067 \caption@ifpackageloaded{fltpage}[1998/10/29 v.0.3]{%
2068   \caption@setbool{needfreeze}{1}%

```

`\FP@helpNote` Original code:

```

\newcommand{\FP@helpNote}[2]{%
  \typeout{FP#1 is inserted on page \pageref{#2}!}}%

2069 \renewcommand\FP@helpNote[2]{%
2070   \begingroup % save \caption@thepage
2071     \caption@pageref{#2}%
2072     \typeout{FP#1 is inserted on page \caption@thepage!}%
2073   \endgroup}%

```

`\FP@floatBegin` Original code:

```

\newcommand{\FP@floatBegin}[1]{%
  \gdef\@capytype{#1}%
  \global\let\FP@savedCaptionCommand\caption%
  \global\let\FP@savedLabelCommand\label%
  \ifthenelse{\equal{\@capytype}{figure}}{
    {\global\let\old@Fnum\fnun@figure}%
    {\global\let\old@Fnum\fnun@table}%
  }
  \let\FP@LabelText\@empty%
  \let\FP@CaptionText\@empty%
  \let\FP@optionalCaptionText\@empty%
  \renewcommand\label[1]{\gdef\FP@LabelText{##1}}%
  \renewcommand\caption[2][{}]{%
    \gdef\FP@optionalCaptionText{##1}\gdef\FP@CaptionText{##2}}%
  \begin{lrbox}{\FP@floatCorpusBOX}%
}

```

```

2074 \renewcommand*\FP@floatBegin[1]{%
2075   \def\@capttype{#1}%
2076   \let\FP@LabelText\@empty
2077   \begin{lrbox}{\FP@floatCorpusBOX}%
2078   \caption@ifFP@prefcap
2079     {\caption@freeze\relax}%
2080     {\def\label##1{\gdef\FP@LabelText{##1}\ignorespaces}%
2081     \caption@freeze*}%

```

\FP@floatEnd **Original code:**

```

\newcommand{\FP@floatEnd}{%
  \end{lrbox}%
  \global\setbox\FP@floatCorpusBOX=\box\FP@floatCorpusBOX
  \stepcounter{FP@\@capttype C}%
  \FP@savedLabelCommand{\FP@positionLabel}%
  \FP@helpNote{\@capttype}{\FP@positionLabel}%
  \FP@float
    {\FP@positionLabel}% location label test
    {\begin{\@capttype}[p!]}
      \usebox{\FP@floatCorpusBOX}%
      \refstepcounter{\@capttype}%
      \ifthenelse{\equal{\FP@LabelText}{\@empty}}{
        {\FP@savedLabelCommand{\expandafter\protect\FP@LabelText}}%
      }{}
    \end{\@capttype}}
  {\addtocounter{\@capttype}{-1}}
  {\begin{\@capttype}[b!]}%
    \ifthenelse{\equal{\FP@guide}{\@empty}}{
      {}{\ifthenelse{\equal{\@capttype}{figure}}{
        {\renewcommand{\fnum@figure}{\old@Fnum\ {\FP@guide}}}%
        {\renewcommand{\fnum@table}{\old@Fnum\ {\FP@guide}}}%
      }{}
      \setlength{\abovecaptionskip}{2pt plus2pt minus 1pt} % length above caption
      \setlength{\belowcaptionskip}{2pt plus2pt minus 1pt} % length below caption
      \FP@separatorCaption%
      \ifthenelse{\equal{\FP@optionalCaptionText}{\@empty}}{
        {\FP@savedCaptionCommand{\expandafter\protect\FP@CaptionText}}%
        {\FP@savedCaptionCommand[\expandafter\protect\FP@optionalCaptionText]%
          {\expandafter\protect\FP@CaptionText}}%
      }{}
    }
  \end{\@capttype}}%
}%

```

```

2082 \renewcommand*\FP@floatEnd{%
2083   \end{lrbox}%
2084   \stepcounter{FP@\@capttype C}%
2085   \caption@label\FP@positionLabel
2086   \FP@helpNote\@capttype\FP@positionLabel
2087   \let\FP@ContinuedFloat\relax
2088   \FP@float
2089     {\FP@positionLabel}% location label test

```

```

2090     {\begin\@capttype[p!]}%
2091         \FP@ContinuedFloat
2092         \usebox\FP@floatCorpusBOX
2093         \caption@ifFPlistcap
2094             {\caption@refstepcounter\@capttype
2095              \expandafter\caption@makecurrent\expandafter\@capttype
2096                  \expandafter{\caption@SClentry}}}%
2097             {\expandafter\captionlistentry\expandafter{\caption@SClentry}}}%
2098         \caption@@anchor\relax
2099         \ifx\FP@LabelText\@empty \else
2100             \expandafter\label\expandafter{\FP@LabelText}%
2101         \fi
2102     \end\@capttype}%
2103 {\let\FP@ContinuedFloat\ContinuedFloat
2104  \caption@ifcompatibility
2105      {\let\caption@ContinuedFloat\@gobble}%
2106      {\caption@clearsetup{ContinuedFloat}%
2107       \caption@clearsetup{continued\@capttype}}}%
2108 {\begin\@capttype[b!]}%
2109     \FP@ContinuedFloat
2110     \let\FP@savedSetfnumCommand\caption@setfnum
2111     \def\caption@setfnum##1{%
2112         \FP@savedSetfnumCommand{##1}%
2113         \ifx\FP@guide\@empty \else
2114             \expandafter\l@addto@macro\csname fnum@##1\endcsname{\ \FP@guide}}%
2115     \fi}%
2116     \setlength\abovecaptionskip{2pt plus 2pt minus 1pt}% length above caption
2117     \setlength\belowcaptionskip{2pt plus 2pt minus 1pt}% length below caption
2118     \caption@setoptions{FP\@capttype}%
2119     \FP@separatorCaption
2120     \caption@ifFPlistcap{}{\caption@set@bool\caption@iflof0}%
2121     \caption@defrost
2122     \end\@capttype}%
2123 }%

2124 \ForFloatingEnvironments{%
2125     \newenvironment{FP#1}{\FP@floatBegin{#1}}{\FP@floatEnd}}%
2126 }%
2127 \let\caption@ifFPlistcap\@undefined
2128 \let\caption@ifFPrefcap\@undefined
2129 }

```

2.19.4 The hyperref package

```

2130 \caption@ifPackageLoaded{hyperref}[2003/11/30 v6.74m]{%
2131     \@ifundefined{hyper@makecurrent}{% hyperref has stopped early
2132         \caption@WarningNoLine{%
2133             Hyperref support is turned off\MessageBreak
2134             because hyperref has stopped early}%
2135     }{%

```

```

2136      \g@addto@macro\caption@prepareslc{\measuring@true}%
\caption@@refstepcounter  We redefine \caption@@refstepcounter so \H@refstepcounter will be
                          used instead of \refstepcounter inside \caption & \captionlistentry.
2137      \renewcommand*\caption@@refstepcounter{\H@refstepcounter}%

\caption@makecurrent      We redefine \caption@makecurrent so a hyperref label will be defined inside
                          \@caption.
                          Note: Will be redefined by \caption@start.
2138      \renewcommand*\caption@makecurrent[2]{%
2139          \caption@makecurrentHref{#1}%
2140          \caption@Debug{hyperref current=\@currentHref}%
2141          \def\@currentlabelname{#2}}%
2142      \newcommand*\caption@makecurrentHref{\hyper@makecurrent}%

\caption@@anchor          We redefine \caption@@anchor so a hyperref anchor will be set inside \@caption.
                          Note: Will be redefined by \caption@start.
2143      \renewcommand\caption@@anchor[1]{%
2144          \caption@Debug{hyperref anchor: \@currentHref}%
2145          % If we cannot have nesting, the anchor is empty.
2146          \ifHy@nesting
2147              \hyper@@anchor{\@currentHref}{#1}%
2148          \else
2149              \Hy@raisedlink{\hyper@@anchor{\@currentHref}{\relax}}#1%
2150              \fi}%
2151      \g@addto@macro\caption@prepareslc{\let\caption@@anchor\@firstofone}%

\caption@@ContinuedFloat  When the hyperref package is used we have the problem that the usage of \ContinuedFloat
                          will create duplicate hyper links – \@currentHref will be the same for the main float
                          and the continued ones. So we have to make sure unique labels and references will be
                          created each time. We do this by extending \theHfigure and \theHtable, so for
                          continued floats the scheme

                              <type> . <type #> \alph{<continued #>}

                          will be used instead of

                              <type> . <type #> .

                          (This implementation follows an idea from Steven Douglas Cochran.)
                          Note: This does not help if the hyperref package option naturalnames=true is set.
2152      \renewcommand*\caption@@ContinuedFloat[1]{%
2153          \ifundefined{theH#1}{}%
2154          \expandafter\l@addto@macro\csname theH#1\endcsname{%
2155              \@alph@c@ContinuedFloat}}}%

```

The hypcap option

`\if@capstart` Like the `hypcap` package we define the switch `\if@capstart`, too.

```
2156 \newif\if@capstart
```

`\caption@start` While the `hypcap` package defines a macro called `\capstrart` our variant is called `\caption@start` and is controlled by the option `hypcap=false/true`.

```
2157 \def\caption@start{\caption@ifhypcap{%
```

Generate the `hyperref` label and set the `hyperref` anchor, usually (if `hypcap=false`) both is done inside `\@caption`.

```
2158 \caption@makestart\@captype
```

```
2159 \caption@startanchor\@currentHref
```

Prevent `\@caption` from generating a new `hyperref` label, use the label we save in `\hc@currentHref` instead. (We also support the `@capstart` flag from the `hypcap` package.)

```
2160 \global\@capstarttrue
```

```
2161 \let\hc@currentHref\@currentHref
```

```
2162 \def\caption@makecurrentHref##1{%
```

```
2163 \global\@capstartfalse
```

```
2164 \global\let\@currentHref\hc@currentHref}%
```

Prevent `\@caption` from generating a `hyperref` anchor since this has already been done.

```
2165 \let\caption@@anchor\@firstofone
```

```
2166 }{}%}
```

`\caption@makestart` `\caption@makestart{<type>}` defines a `hyperref` anchor inside `\caption@start`. Since we offer `\ContinuedFloat` the float counter can change between ‘now’ and `\caption`, i.e., we simply don’t know the figure or table counter yet and therefore we are not able to generate the ‘right’ `hyperref` label. Two different solutions of this problem came into my mind:

1. I could use the aux file for this purpose.

-or-

2. I set `hypertextnames=false` locally. Furthermore I use `#1.caption.<counter>` (instead of `#1.<counter>`) as naming scheme for `\@currentHref` to avoid conflicts with other hyper links which are generated with `hypertextnames=true`.

The first idea has the advantage that the ‘right’ anchor name will be generated, but one needs an additional `LATEX` run if figures or tables will be inserted or removed.

The second idea has the advantage that it’s very easy to implement, but has some side-effects, e.g. the anchor names don’t follow the figure or table label names anymore.

Since I’m lazy I implemented the second idea, maybe I will revise this later on.

```
2167 \newcommand*\caption@makestart[1]{%
```

```
2168 \begingroup
```

```
2169 \Hy@hypertextnamesfalse
```

```
2170 % \gdef\@currentHlabel{}%
```

```

2171         \hyper@makecurrent{#1.caption}%
2172     \endgroup
2173     \caption@Debug{hycap start=\@currentHref}}%

\caption@startanchor \caption@startanchor{<Href>} sets a hyperref anchor inside \caption@start.
This code was taken from the hycap package[?] and adapted.
Note: Since \hyper@@anchor{<Href>}{\relax} can cause a change from vertical mode to
horizontal mode (design flaw in hyperref package!?), and since the workaround \let\leavevmode\relax
which can be found in the hycap package is not always sufficient (for example with “Di-
rect pdfmark support” and breaklinks=true), we use \caption@anchor instead of
\hyper@@anchor here.

2174     \newcommand*\caption@startanchor[1]{%
2175         \caption@Debug{hycap anchor: #1}%
2176         \ifvmode\begin{group}
2177             \@tempdima\prevdepth
2178             \nointerlineskip
2179             \vspace*{-\caption@hycap@space}%
2180             \caption@anchor{#1}%
2181             \vspace*{\caption@hycap@space}%
2182             \prevdepth\@tempdima
2183         \endgroup\else
2184             \caption@Warning{Internal Warning: Vertical TeX mode expected}%
2185             \caption@anchor{#1}%
2186         \fi}%

\caption@anchor \caption@anchor{<Href>} sets a hyperref anchor.

2187     \newcommand*\caption@anchor[1]{%
2188         \ifmeasuring@ \else
2189             \caption@raisedlink{\hyper@anchorstart{#1}\hyper@anchorend}%
2190         \fi}%

Note: Since \Hy@raisedlink change \@tempdima we surrounded it by \ifvmode, sup-
pressing “LaTeX Warning: Float too large for page by 1.0pt” in sideways
floats. (This will not be necessary with future versions of hyperref.)

2191     \ifx\HyperRaiseLinkLength\@tempdima
2192         \def\caption@raisedlink#1{\ifvmode#1\else\Hy@raisedlink{#1}\fi}%
2193     \else
2194         \let\caption@raisedlink\Hy@raisedlink
2195     \fi

\caption@@start Will be used by \caption@freezeHref. Apart from that we issue a warning if we
expect a saved hyperref label coming from \caption@start, but there isn’t any.

2196     \def\caption@@start{%
2197         \ifundefined{hc@currentHref}{%
2198             \caption@Warning{%
2199                 The option ‘hycap=true’ will be ignored for this\MessageBreak
2200                 particular \string\caption}}}%

```

`\caption@freezeHref` Suppress `\caption@start` from generating a `hyperref` label and setting a `hyperref` anchor. Instead if `\@caption` generates a `hyperref` label, it will be stored in `\caption@currentHref`. Furthermore we need to redefine `\caption@setfloatcapt` so no `hyperref` anchor will be placed in `\@caption`.

```

2201 \def\caption@freezeHref{%
2202   \let\caption@ORI@start\caption@start
2203   \def\caption@start{\let\caption@start\caption@ORI@start}%
2204 %
2205   \let\caption@ORI@@start\caption@@start
2206 %
2207   \l@addto@macro\subcaption@settypehook{%
2208     \let\caption@@start\caption@ORI@@start}%
2209 %
2210   \global\let\caption@currentHref\@undefined
2211   \def\caption@@start{\global\let\caption@currentHref\@currentHref}%
2212 %
2213   \let\caption@ORI@setfloatcapt\caption@setfloatcapt
2214   \renewcommand*\caption@setfloatcapt{%
2215     \ifx\caption@currentHref\@undefined \else
2216       \let\caption@@anchor\@firstofone
2217     \fi
2218     \caption@ORI@setfloatcapt}}%

```

`\caption@defrostHref` If there is a frozen `\@currentHref`, we set the `hyperref` anchor here.

```

2215 \def\caption@defrostHref{%
2216   \ifx\caption@currentHref\@undefined \else
2217     \caption@startanchor\caption@currentHref
2218   \global\let\caption@currentHref\@undefined
2219   \fi}%

```

`\float@makebox` Do our own redefinition of `\float@makebox`, if it was redefined by the `hyperref` package.

```

2220 \@ifundefined{HyOrg@float@makebox}{}{%
2221   \caption@Debug{%
2222     Redefining \noexpand\float@makebox (again)\@gobble}%
2223   \let\caption@ORI@float@makebox\float@makebox % save for compatibility mode
2224   \renewcommand\float@makebox[1]{%
2225     \HyOrg@float@makebox{#1}\relax \caption@defrostHref}}%
2226   }%
2227 }{}

```

2.19.5 The `hycap` package

```

2228 \caption@IfPackageLoaded{hycap}{% v1.0
2229   \ifx\caption@start\relax \else % hyperref hasn't stopped early

```

If the `hycap` package was loaded, we give up our own hyperlink placement algorithm and give the control over the placement to the `hycap` package instead.

`\capstart` We do this simply by mapping `\capstart` to `\caption@start`, although our code does not behave exactly like the original one: The original `\capstart` has an effect on

the next `\caption` only but our version affects *all* `\captions` in the same environment, at least unless a new `\capstart` will be placed.

```
2230 \let\caption@ORI@capstart\capstart % save for compatibility mode
2231 \let\capstart\caption@start
2232 \let\caption@start\relax
2233 \let\caption@@start\relax
```

`\caption@hypcapSPACE` Furthermore we map our `\caption@hypcapSPACE` to `\hypcapSPACE` offered by the `hyrcap` package.

```
2234 \caption@set@bool\caption@ifhyrcap1%
2235 \renewcommand*\caption@hyrcapSPACE{\hyrcapSPACE}%
2236 \fi}{}
```

2.19.6 The listings package

```
2237 \caption@IfPackageLoaded{listings}[2004/02/13 v1.2]{%
```

`\lst@MakeCaption` To support the `listings` package we need to redefine `\lst@MakeCaption` so the original stuff is nested with `\caption@begin` and `\caption@end`.

```
2238 \let\caption@ORI@lst@MakeCaption\lst@MakeCaption
2239 \def\lst@MakeCaption#1{%
2240 \caption@setoptions{lstlisting}%
```

If the `position=` is set to `auto`, we take over the `captionpos=` setting from the `listings` package. Note that we won't do this otherwise, so `listings` settings like `abovecaptionskip=0pt`, `belowcaptionskip=10pt`, `captionpos=t` will *not* cause different outputs with or without the `caption` package loaded.

```
2241 \caption@setautoposition{#1}%
2242 \caption@begin{lstlisting}%
2243 \caption@ORI@lst@MakeCaption{#1}%
2244 \caption@end}%
```

`\ext@lstlisting` Since the `listings` package do not define `\ext@lstlisting`, but we needed it when `\captionof{lstlisting}` will be done by the end user, we define it here.

```
2245 \providecommand*\ext@lstlisting{lol}%
2246 }{}
```

2.19.7 The longtable package

`\LTcaption` `\LTcaption` is preset to `table`.

```
2247 \providecommand*\LTcaption{table}

2248 \caption@IfPackageLoaded{longtable}[1995/05/24 v3.14]{%
2249 \RequirePackage{ltcaption}[2007/09/01]%
2250 \let\LT@@makecaption\@undefined
```

```

\LT@array We redefine \LT@array here to get \captionsetup{<options>} working inside
longtables.
Note: Since the hyperref package patches \LT@array as well and since this only works
with the original definition of \LT@array, we have to do this after the hyperref package,
i.e. \AtBeginDocument.
2251 \caption@AtBeginDocument{%
2252 \let\caption@ORI@LT@array\LT@array
2253 \renewcommand*\LT@array{%
\captionsetup for longtable:
2254 \global\let\caption@opt@@longtable\@undefined
2255 \def\captionsetup##1{\noalign{\LT@captionsetup{##1}}}%
2256 \def\LT@captionsetup##1{%
2257 \captionsetup\@gobble[@longtable]{##1}%
2258 \global\let\caption@opt@@longtable\caption@opt@@longtable}%
\captionabove & \captionbelow for longtable: (KOMA-Script document class)
2259 \def\captionabove\true{\LT@captionsetup{position=t}}%
2260 \def\captionabove\false{\LT@captionsetup{position=b}}%
\captionlistentry for longtable:
2261 \def\captionlistentry##1{\noalign{%
2262 \caption@listentry[\LT@captiontype]{##1}}}%
\ContinuedFloat for longtable:
(Commented out, since it's not deeply tested and quite useless anyway)
Note: hyperref versions < v6.76j uses 2x \hyper@makecurrent
2263 % \caption@ifhyppcap{%
2264 % \let\caption@ORI@hyper@makecurrent\hyper@makecurrent
2265 % \def\hyper@makecurrent##1{%
2266 % \let\hyper@makecurrent\caption@ORI@hyper@makecurrent
2267 % \caption@makestart{##1}%
2268 % \let\Hy@LT@currentHlabel\@currentHlabel
2269 % \let\Hy@LT@currentHref\@currentHref
2270 % \def\hyper@makecurrent####1{%
2271 % \let\@currentHlabel\Hy@LT@currentHlabel
2272 % \let\@currentHref\Hy@LT@currentHref}}%
2273 % \let\caption@ORI@ContinuedFloat\ContinuedFloat
2274 % \def\ContinuedFloat{\noalign{%
2275 % \gdef\caption@setContinuedFloat{%
2276 % \let\caption@resetContinuedFloat\@gobble}%
2277 % \def\caption@setoptions####1{%
2278 % \g@addto@macro\caption@setContinuedFloat{%
2279 % \caption@setoptions{####1}}}%
2280 % \let\@captiontype\LT@captiontype
2281 % \caption@ORI@ContinuedFloat}}%
2282 % }{%
2283 % \def\ContinuedFloat{\noalign{%
2284 % \caption@Error{%
2285 % \noexpand\ContinuedFloat inside longtables\MessageBreak
2286 % is only available with 'hyppcap=true'}}}%

```

```

2287 %      }%
2288 %      \global\let\caption@setContinuedFloat\@empty
2289      \def\ContinuedFloat{\noalign{%
2290          \caption@Error{\noexpand\ContinuedFloat outside float}}}%
2291      \caption@ORI@LT@array}}%

```

`\LT@c@ption` The original implementation:

```

\def\LT@c@ption#1[#2]#3{%
  \LT@makecaption#1\fnun@table{#3}%
  \def\@tempa{#2}%
  \ifx\@tempa\@empty\else
    {\let\\\space
     \addcontentsline{lot}{table}{\protect\numberline{\thetable}{#2}}}%
  \fi}

```

Our implementation uses `\LTcapttype` instead of `{table}`:

```

2292 \def\LT@c@ption#1[#2]#3{%
2293     \LT@makecaption#1{\csname fnun@\LTcapttype\endcsname}{#3}%
2294     \def\@currentlabelname{#2}%
2295     \caption@addcontentsline\LTcapttype{#2}}%

```

`\LT@makecaption` `\LT@makecaption{<cmd>}{<label>}{<text>}`

The original definition:

```

\def\LT@makecaption#1#2#3{%
  \LT@mcol\LT@cols c{\hbox to\z@{\hss\parbox[t]\LTcapwidth{%
    % Based on article class "\@makecaption", "#1" is "\@gobble" in star
    % form, and "\@firstofone" otherwise.
    \sbox\@tempboxa{#1{#2: }#3}%
    \ifdim\wd\@tempboxa>\hsize
      #1{#2: }#3%
    \else
      \hbox to\hsize{\hfil\box\@tempboxa\hfil}%
    \fi
    \endgraf\vskip\baselineskip}%
  \hss}}%

```

Our definition:

```

2296 \renewcommand\LT@makecaption[3]{%
2297     \caption@LT@make{%

```

We set `\ifcaption@star` according to the 1st argument.

```

2298     \caption@startrue#1\caption@starfalse

```

If `\LTcapwidth` is not set to its default value 4in we assume that it shall overwrite our own setting. (But `\captionsetup[longtable]{width=...}` will overwrite `\LTcapwidth`.)

```

2299     \caption@settype*\LTcapttype

```

```

2300     \ifdim\LTcapwidth=4in \else
2301         \setcaptionwidth\LTcapwidth
2302     \fi
2303     \caption@setoptions{longtable}%
2304 %     \caption@setContinuedFloat
2305     \caption@setoptions{@longtable}%

```

position=auto is a bad idea for longtables, but we do our very best. This works quite well for captions inside the longtable contents, but not for captions inside the longtable (end)foot.

Note: This should be ‘top’ if unclear!

```

2306     \caption@setautoposition{\ifcase\LT@rows t\else b\fi}%
2307     \caption@resetContinuedFloat\LTcaptype
2308     \caption@begin\LTcaptype
2309     \caption@normalsize

```

The following skip has the purpose to correct the height of the `\parbox[t]`. Usually it’s the height of the very first line, but because of our extra skips (`\abovecaptionskip` and `\belowcaptionskip`) it’s always 0pt.

(A different idea would be typesetting the first skip outside the longtable column with `\noalign{\vskip...}`, but this means we have to move `\caption@begin` to some other place because it does not work in tabular mode. And at the moment I have no idea on how to do this in an elegant way...)

```

2310     \vskip-\ht\strutbox

```

The following code should look familiar. We do our skips and use `\caption@@make` to typeset the caption itself.

```

2311     \caption@iftop{\vskip\belowcaptionskip}{\vskip\abovecaptionskip}%
2312     \caption@@make{#2}{#3}\endgraf
2313     \caption@iftop{\vskip\abovecaptionskip}{\vskip\belowcaptionskip}%
2314     \caption@end}}%
2315 }{}

```

2.19.8 The picinpar package

```

2316 \caption@IfPackageLoaded{picinpar}{%

```

`\figwindow` The `picinpar` package comes with its own caption code (`\wincaption`, `\@wincaption`,
`\tabwindow` `\@makewincaption`, ...) so we redefine `\figwindow` & `\tabwindow` to use
`\caption` instead.

```

2317 \long\def\figwindow[#1,#2,#3,#4] {%
2318     \caption@window{figure}%
2319     \caption@setoptions{figwindow}%
2320     \begin{window}[#1,#2,{#3},\caption@wincaption{#4}] }%
2321 \long\def\tabwindow[#1,#2,#3,#4] {%
2322     \caption@window{table}%
2323     \caption@setoptions{tabwindow}%
2324     \begin{window}[#1,#2,{#3},\caption@wincaption{#4}] }%

```

`\caption@window` Beside calling `\caption@settype` we redefine `\caption@boxrestore` (as in floatflt & picins package support) and `\@makecaption` (as in float package support) here.

```

2325 \newcommand*\caption@window[1]{%
2326   \let\caption@boxrestore\@parboxrestore
2327   \let\@makecaption\caption@@make
2328   \caption@setautoposition b%
2329   \caption@settype{#1}%
2330   \caption@clearmargin}%

```

`\caption@wincaption` This one finally typesets the caption using `\caption`.

```

2331 \newcommand\caption@wincaption[1]{%

```

This will be done twice for every figwindow & tabwindow caption – on the first run `\picwd` is 0pt, on the second run `\picwd` is `\hsize`.

```

2332   \ifdim\picwd=\z@
2333     \let\caption@makecurrent\@gobbletwo
2334     \let\caption@@start\relax
2335     \caption@prepareslc
2336   \fi

```

The argument #1 could contain simply the caption text (e.g. A figure caption), but it could also contain an optional argument, the *⟨lst.entry⟩* (e.g. [An entry to the LOF]{A figure caption}). Therefore we have to test if #1 begins with [or not; furthermore we support a starred variant – as in `\caption*` – so we test for *, too.

```

2337   \edef\@tempa{\expandafter\noexpand\@car#1\@nil}%
2338   \if\@tempa*%
2339     \let\@tempa\@firstofone
2340   \else\if\@tempa[%
2341     \let\@tempa\@firstofone
2342   \else
2343     \let\@tempa\@empty
2344   \fi\fi
2345   \expandafter\caption\@tempa{#1}}%
2346 }{}

```

2.19.9 The picins package

`\piccaptiontype` `\piccaptiontype{⟨type⟩}`

We offer this macro for changing the *⟨type⟩* of the caption, so the user doesn't have to redefine `\@capytype`, as proposed in the picins documentation.

Note: We define this macro here so it can be used in the preamble of the document, even when the caption package was loaded prior to the picins package.

```

2347 \newcommand*\piccaptiontype[1]{\def\@piccaptiontype{#1}}
2348 \caption@ifpackageloaded{picins}{%

```

Initial set `\@piccaptiontype` and undefine `\@capttype` which was set to figure by the `picins` package.

```
2349 \@ifundefined{@piccaptiontype}{%
2350   \caption@iftype{%
2351     \let\@piccaptiontype\@capttype
2352   }{%
2353     \def\@piccaptiontype{figure}%
2354   }%
2355 }{}%
2356 \let\@capttype\@undefined
```

`\piccaption` The original code:

```
\def\piccaption{\@ifnextchar [{\@piccaption}{\@piccaption[]}}
```

Our code uses `\caption@star` so `\piccaption*` works, and `\caption@dblarg` so `\piccaption{}` works correctly.

```
2357 \def\piccaption{\caption@star\relax{\caption@dblarg\@piccaption}}%
```

`\make@piccaption` The original code:

```
\def\make@piccaption{%
[... ]
\setbox\@TEXT=\vbox{\hsize\hsiz@\caption[\sh@rtf@rm]{\capti@nt@xt}}%
}
```

In our code we have to correct several things:

1. `\@capttype` must be defined, since we have removed the global definition.
2. We use `\caption@setoptions{parpic}` so `\captionsetup[parpic]{...}` is supported.
3. `\linewidth` must be set correctly. Usually this is done by `\@parboxrestore` inside `\@caption`, but since we use `\@caption@boxrestore` we have to map this to `\@parboxrestore` instead.
4. The two arguments of `\caption(\sh@rtf@rm & \capti@nt@xt)` should be expanded on first level so `\caption[] {...}` and `\caption[...]{}` work correctly.

```
2358 \let\caption@ORI@make@piccaption\make@piccaption
2359 \def\make@piccaption{%
2360   \let\caption@ORI\caption
2361   \long\def\caption[##1]##2{%
2362     \caption@freezeHref % will be defrosted in \ivparpic
2363     \caption@settype\@piccaptiontype
2364     \ifnum\c@piccaptionpos>2\relax
2365       \caption@clearmargin
```

```

2366     \else
2367         \captionwidth\z@ % do not use "width=" setting
2368     \fi
2369     \caption@setoptions{parpic}%
2370     \let\caption@boxrestore\@parboxrestore
2371     \caption@setautoposition b%
2372     \expandafter\expandafter\expandafter\caption@ORI
2373     \expandafter\expandafter\expandafter[%
2374     \expandafter\expandafter\expandafter{%
2375     \expandafter##1\expandafter}\expandafter]\expandafter{##2}}%
-or- \begingroup
    \toks0\expandafter{##1}\toks2\expandafter{##2}
    \edef\x{\endgroup
        \noexpand\caption@ORI[{\the\toks0}]{\the\toks2}}
    \x
-or- \edef\x{%
    \noexpand\caption@ORI[{\unexpanded\expandafter{##1}}]%
        {\unexpanded\expandafter{##2}}}
    \x
2376     \caption@ORI@make@piccaption
2377     \let\caption\caption@ORI}%

```

`\ivparpic` We need to set our `hyperref` anchor here. Not bullet-proof since we have to redefine `\noindent` here!

```

2378 \let\caption@ORI@ivparpic\ivparpic
2379 \def\ivparpic(#1,#2)(#3,#4)[#5][#6]#7{%
2380     \let\caption@ORI@noindent\noindent
2381     \def\noindent{%
2382         \caption@defrostHref
2383         \let\noindent\caption@ORI@noindent
2384         \noindent}%
2385     \caption@ORI@ivparpic(#1,#2)(#3,#4)[#5][#6]{#7}%
2386     \let\noindent\caption@ORI@noindent}%
2387 }%
2388 \let\piccaptiontype\@undefined
2389 }

```

2.19.10 The rotating package

```

2390 \caption@ifPackageLoaded{rotating}[1995/08/22 v2.10]{%
\rotcaption Make \rotcaption* work.
2391 \def\rotcaption{\let\@makecaption\@makerotcaption\caption}%
2392 % \let\@rotcaption\@undefined
\rotcaptionof Make \rotcaptionof(*) work.
2393 \def\rotcaptionof{%
2394     \caption@teststar\caption@of{\rotcaption*}\rotcaption}%

```

`\@makerotcaption` Original (bugfixed) code:

```
\long\def\@makerotcaption#1#2{%
  \setbox\@tempboxa\hbox{#1: #2}%
  \ifdim \wd\@tempboxa > .8\vsiz
    \rotatebox{90}{%
      \begin{minipage}{.8\textheight}#1: #2\end{minipage}%
    }%\par % <== \par removed (AR)
  \else%
    \rotatebox{90}{\box\@tempboxa}%
  \fi
  \nobreak\hspace{12pt}% <== \nobreak added (AR)
}
```

Our version emulates this behavior, but if `width=` is set, the rotated caption is always typeset as minipage. (Note that `margin=` is not supported here.)

```
2395 \long\def\@makerotcaption#1#2{%
2396   \ifdim\captionwidth=\z@
2397     \setcaptionwidth{.8\textheight}%
2398     \caption@slc{#1}{#2}{.8\vsiz}%
2399     \let\caption@makerot\caption@@make
2400     \caption@clearmargin
2401 %     \long\def\captionbox##1##2{\hbox{\hsize=.8\textheight\relax##2}}%
2402 %     (not needed because \rotatebox uses an \hbox anyway)
2403     \let\captionbox\@secondoftwo%
2404     \caption@set@bool\caption@ifslc0% been there, done that
2405   \fi
2406   \rotatebox{90}{\caption@makerot{#1}{#2}}%
2407   \nobreak\hspace{12pt}}%
2408 \newcommand\caption@makerot[2]{%
2409   \begin{minipage}\captionwidth\caption@@make{#1}{#2}\end{minipage}}%
2410 \ForFloatingEnvironments{%
2411   \newenvironment{sideways#1}{\@rotfloat{#1}}{\end@rotfloat}%
2412   \newenvironment{sideways#1*}{\@rotdblfloat{#1}}{\end@rotdblfloat}}%
2413 {} }
```

2.19.11 The sidecap package

```
2414 \caption@ifPackageLoaded{sidecap}[1999/05/11 v1.4d]{%
2415   \caption@setbool{needfreeze}{1}%
```

`\SC@caption` First of all, we let `sidecap` use a current definition of `\caption`.
(This is only required for version 1.5d of the `sidecap` package.)

```
2416 \caption@AtBeginDocument{\let\SC@caption=\caption}%
```

`\SC@zfloat` This macro will be called at the start of the environment, here is a good opportunity to do some adaptations to `\caption` and `\captionsetup`.

```

2417 \let\caption@ORI@SC@zfloat\SC@zfloat
2418 \def\SC@zfloat#1#2#3[#4]{%

```

First we use the original definition, but save & restore `\caption` so `\caption@freeze` will work correctly.

```

2419 \let\caption@ORI\caption
2420 \caption@ORI@SC@zfloat{#1}{#2}{#3}[#4]%
2421 \let\caption\caption@ORI

```

Since the `sidecap` package uses our `\caption` code outside the environment the regular `\captionsetup` will not work. So we need a special version here which saves the given argument list which will be executed later on. Furthermore we need to make `\caption*` work.

```

2422 \caption@settype*{#2}%
2423 \caption@freeze*}%

```

`\endSC@FLOAT` This macro will be called at the end of the environment, here we need to setup our stuff before the `sidecap` package actually typesets its caption.

```

2424 \let\caption@ORI@endSC@FLOAT\endSC@FLOAT
2425 \def\endSC@FLOAT{%

```

Note: `\@cptype` isn't defined here, this will be done inside the original definition of `\endSC@FLOAT`. But `\SC@cptype` is defined and can be used here, if needed.

```

2426 \let\caption@ORI@settype\caption@settype
2427 \def\caption@settype##1{% will be done in \@xfloat
2428 \caption@ORI@settype*{##1}% do not change \@currentlabel
2429 \caption@setSC@justify
2430 %% \caption@setoptions{SCfloat}%
2431 \caption@setoptions{SC\@cptype}%
2432 \caption@start}%

```

Before we can typeset the caption we need to set the margin to zero because any extra margin would only be disturbing here.

(We don't need to take care about the caption position because the `sidecap` package set both `\abovecaptionskip` and `\belowcaptionskip` to a skip of zero anyway.)

Furthermore `\SC@justify` will override the caption justification, if set. The usage of `\SC@justify` differs from version to version of the `sidecap` package:

Version 1.4: `\SC@justify` is not defined

Version 1.5: `\SC@justify` is `\relax` when not set

Version 1.6: `\SC@justify` is `\@empty` when not set

```

2433 \def\caption@setSC@justify{%
2434 \caption@clearmargin
2435 \@ifundefined{SC@justify}{}{%
2436 \ifx\SC@justify\@empty \else
2437 \let\caption@hj\SC@justify
2438 \let\SC@justify\@empty
2439 \fi}%

```

Make the original definition of `\endSC@FLOAT` to use our caption stuff instead of its own.

Note: At this point the `sidecap` definition of `\caption` is valid, not the regular one!

```

2440 \let\caption\SC@orig@caption
2441 \def\SC@orig@caption[##1]##2{\caption@defrost}%

Finally we call the original definition of \endSC@FLOAT.

2442 \caption@setSC@justify % for compatibility mode
2443 \caption@ORI@endSC@FLOAT}%

2444 \ForFloatingEnvironments{%
2445 \newenvironment{SC#1}%
2446 {\SC@float[\@nameuse{SC@#1@vpos}]{#1}}{\endSC@float}%
2447 \newenvironment{SC#1*}%
2448 {\SC@dblfloat[\@nameuse{SC@#1@vpos}]{#1}}{\endSC@dblfloat}}%
2449 {} }
```

2.19.12 The subfigure package

```

2450 \caption@ifPackageLoaded{subfigure}[2002/01/23 v2.1]{%
\sif@ifpositiontop If the subfigure package is loaded, we map \sf@ifpositiontop to \iffiguretopcap
resp. \iftabletopcap, so the subfigure v2.1 options figbotcap etc. will still work.

2451 \def\sif@ifpositiontop{%
2452 \ifx\@capytype\undefined
2453 \expandafter\@gobbletwo
2454 \else\ifx\@capytype\relax
2455 \expandafter\expandafter\expandafter\@gobbletwo
2456 \else
2457 \expandafter\expandafter\expandafter\sif@if@position@top
2458 \fi\fi}

2459 \def\sif@if@position@top{%
2460 \@ifundefined{if\@capytype topcap}%
2461 {\@gobbletwo}%
2462 {\@nameuse{if\@capytype topcap}}%
2463 \expandafter\@firstoftwo
2464 \else
2465 \expandafter\@secondoftwo
2466 \fi}}

2467 {} }
```

2.19.13 The supertabular package

```

2468 \caption@ifPackageLoaded{supertabular}[2002/07/19 v4.1e]{%
\tablecaption Make \topcaption* and \bottomcaption* work.

2469 \let\caption@ORI@tablecaption\tablecaption
2470 \def\tablecaption{\caption@star\relax\caption@ORI@tablecaption}%

```

`\ST@caption` **The original code:**

```

\long\def\ST@caption#1[#2]#3{\par%
  \addcontentsline{\csname ext@#1\endcsname}{#1}%
    {\protect\numberline{%
      \csname the#1\endcsname}{\ignorespaces #2}}
  \begingroup
    \@parboxrestore
    \normalsize
    \if@topcaption \vskip -10\p@ \fi
    \@makecaption{\csname fnum@#1\endcsname}{\ignorespaces #3}\par
    \if@topcaption \vskip 10\p@ \fi
  \endgroup}

2471 \long\def\ST@caption#1[#2]#3{\par%
2472   \caption@settype*{#1}%
2473   \caption@setoptions{supertabular}%

  The position= setting will be overwritten by the supertabular package: If \topcaption
  was used, the position will be top automatically, bottom otherwise.

2474   \def\caption@fixposition{%
2475     \caption@setposition{\if@topcaption t\else b\fi}}%

2476   \caption@beginex{#1}{#2}{#3}%
2477   \caption@boxrestore
2478   \caption@normalsize
2479   \@makecaption{\csname fnum@#1\endcsname}{\ignorespaces #3}\par
2480   \caption@end}%

2481 }{%
2482 \let\caption@setSTposition\@undefined
2483 }
```

2.19.14 The threeparttable package

```

2484 \caption@ifpackageloaded{threeparttable}[2003/06/13 v3.0]{%

\threeparttable    Unfortunately \caption is not set when \TPT@common will be used, so we have to
                   redefine \threeparttable and \measuredfigure instead.

2485   \let\caption@ORI@threeparttable\threeparttable
2486   \renewcommand*\threeparttable{%
2487     \caption@settype{table}%
2488     \caption@setposition a% ?
2489     \caption@clearmargin
2490     \caption@setoptions{threeparttable}%
2491     \caption@ORI@threeparttable}%

\measuredfigure    Same here...

2492   \let\caption@ORI@measuredfigure\measuredfigure
2493   \renewcommand*\measuredfigure{%
```

```

2494 \caption@settype{figure}%
2495 \caption@setposition a% ?
2496 \caption@clearmargin
2497 \caption@setoptions{measuredfigure}%
2498 \caption@ORI@measuredfigure}%

```

\TPT@caption **The original code:**

```

\def\TPT@caption#1[#2]#3{\gdef\TPT@docapt
{\par\global\let\TPT@docapt\@undefined \TPT@LA@caption{#1}[{#2}]}%
{\strut\ignorespaces#3\ifhmode\unskip\@finalstrut\strutbox\fi}}%
\ifx\TPT@hsize\@empty \let\label\TPT@gatherlabel \abovecaptionskip\z@skip
\else \TPT@docapt \fi \ignorespaces}

2499 \def\TPT@caption#1[#2]#3{%
2500 \gdef\TPT@docapt{%
2501 \global\let\TPT@docapt\@undefined
2502 \caption@setautoposition\caption@TPT@position
2503 \TPT@LA@caption{#1}[{#2}]{#3}}%
2504 \ifx\TPT@hsize\@empty
2505 \let\label\TPT@gatherlabel % Bug: does not work for measuredfigures
2506 \gdef\caption@TPT@position{t}%
2507 \g@addto@macro\TPT@docapt\caption@TPT@eatvskip
2508 \else
2509 \def\caption@TPT@position{b}%
2510 \TPT@docapt
2511 \fi
2512 \ignorespaces}%

2513 %\newcommand*\caption@TPT@eatvskip{\vskip-.2\baselineskip}%
2514 \def\caption@TPT@eatvskip#1\vskip{#1\@tempdima=}%

2515 }{}

```

2.19.15 The wrapfig package

```

2516 \caption@IfPackageLoaded{wrapfig}{% ver 3.3 (Oct 12, 1999)

\float@ifstyle \float@ifstyle{<type>}{<if-clause>}{<else-clause>}
(see float package support for details)

2517 \providecommand*\float@ifstyle[1]{%
2518 \expandafter\ifx\csname fst@#1\endcsname\relax
2519 \expandafter\@secondoftwo
2520 \else
2521 \expandafter\@firstoftwo
2522 \fi}%

```

\caption@restylewrapfloat This one redefines the wrap#1 environment, e.g. wrapfigure. Our code uses \caption@setoptions{wrapfigure} so \captionsetup[wrapfigure]{...} will work.

But first we check if our redefinition was already done, this could happen inside `\float@restyle` when the `wrapfig` support of the `float` package was not installed successfully, so it has not redefined `\wrap#1` there.

```

2523 \newcommand*\caption@restylewrapfloat[1]{%
2524   \expandafter\ifx\csname caption@OUR@wrap#1\expandafter\endcsname
2525     \csname wrap#1\endcsname
2526   \caption@Error{%
2527     For a successful cooperation of the 'wrapfig' package\MessageBreak
2528     with the 'float' package you should load the 'wrapfig'\MessageBreak
2529     package *after* the 'float' package}%
2530   \else
2531     \expandafter\let\csname caption@ORI@wrap#1\expandafter\endcsname
2532       \csname wrap#1\endcsname
2533     \@namedef{wrap#1}{\caption@wrapfloat{#1}}%
2534     \expandafter\let\csname caption@OUR@wrap#1\expandafter\endcsname
2535       \csname wrap#1\endcsname
2536   \fi}%

```

`\caption@wrapfloat`

```

2537 \newcommand*\caption@wrapfloat[1]{%
2538   \caption@settype*{#1}%
2539   \float@ifstyle{#1}{%
2540     \ifx\WF@floatstyhook\@undefined
2541       \caption@Error{%
2542         For a successful cooperation of the 'wrapfig' package\MessageBreak
2543         with the 'float' package you should use at least\MessageBreak
2544         'wrapfig' version 3.6}%
2545     \else
2546       \float@dostyle{#1}%
2547     \fi}}%
2548   \caption@clearmargin
2549   %%% \caption@setoptions{wrapfloat}%
2550   \caption@setoptions{wrap#1}%
2551   \@nameuse{caption@ORI@wrap#1}}%

```

Now we redefine the `wrapfig` environments we know about.

If someone has placed a `\newfloat` right between `\usepackage{wrapfig}` and `\usepackage{caption}` (or loads the `caption` package first, so all these patches will be done with `\AtBeginDocument`) we have bad luck since the `float` package do not offer a list of (re)styled floats. (This would finally lead to an error in `\caption@setfloatcapt`.)

```

2552 \caption@restylewrapfloat{figure}%
2553 \caption@restylewrapfloat{table}%
2554 \ForFloatingEnvironments{%
2555   \newenvironment{wrap#1}{\wrapfloat{#1}}{\endwrapfloat}%
2556   \caption@restylewrapfloat{#1}}%
2557 \ifx\WF@floatstyhook\@undefined \else % wrapfig v3.6

```

`\float@restyle` If the `wrapfig` package v3.6 is used, we patch `\float@restyle` (if defined), too, so new or restyled floats will be handled correctly, too.

```
2558 \ifundefined{float@restyle}{}{%
2559 \toks@=\expandafter{\float@restyle{#1}% (env may or may not be defined)
2560 \caption@restylewrapfloat{#1}}%
2561 \edef\@tempa{\def\noexpand\float@restyle##1{\the\toks@}}%
2562 \@tempa}% perform redefinitions
```

`\wrapfloat` An additional check of the package load order: If both, neither the `wrapfig` package nor the `caption` package haven't catch `\float@restyle`, we finally splash down at `\wrapfloat`.

```
2563 \let\caption@ORI@wrapfloat\wrapfloat
2564 \def\wrapfloat#1{%
2565 \float@ifstyle{#1}{%
2566 \caption@Error{%
2567 For a successful cooperation of the 'wrapfig' package\MessageBreak
2568 with the 'float' package you should load the 'wrapfig'\MessageBreak
2569 package *right after* the 'float' package}}}%
2570 \caption@ORI@wrapfloat{#1}}%

2571 \fi % wrapfig v3.6
```

`\WF@rapt` We place our hyperref anchor here.
Original code:

```
\def\WF@rapt[#1]#2{% final two args: #1 = overhang, #2 = width,
\gdef\WF@ovh{#1}% hold overhang for later, when \width is known
\global\setbox\WF@box\top\bggroup \setlength\hsize{#2}%
\ifdim\hsize>\z@ \@parboxrestore \else
\setbox\z@\hbox\bggroup \let\wf@@caption\caption \let\caption\wf@caption
\ignorespaces \fi}
```

Our code:

```
2572 \def\WF@rapt[#1]#2{% final two args: #1 = overhang, #2 = width,
2573 \gdef\WF@ovh{#1}% hold overhang for later, when \width is known
2574 \global\setbox\WF@box\top\bggroup \setlength\hsize{#2}%
2575 \caption@start
2576 \ifdim\hsize>\z@ \@parboxrestore \else
2577 \setbox\z@\hbox\bggroup \let\wf@@caption\caption \let\caption\wf@caption
2578 \ignorespaces \fi}%

2579 {} }
```

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